



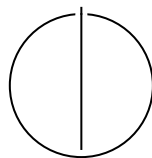
SCHOOL OF COMPUTATION, INFORMATION
AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

Image Classification for Nazi Symbols

Zhiwei Zhang





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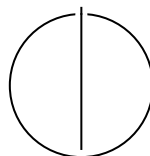
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**Bildklassifikation zur Erkennung
nationalsozialistischer Symbole**

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Submission Date:	15.07.2025



I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 15.07.2025

Zhiwei Zhang

Acknowledgments

First and foremost, I would like to express my deepest gratitude to my wife for her unwavering support, patience, and encouragement throughout the entire course of this thesis. Her belief in me made all the difference during challenging times.

I would like to sincerely thank Professor Jürgen Pfeffer for giving me the opportunity to pursue this research topic and for fostering an academic environment that encourages intellectual curiosity and critical thinking.

I would also like to thank Dr. Angelina Voggenreiter for her kind assistance in helping me explore and identify potential thesis topics within the chair.

Special thanks go to my colleagues and my manager at Allianz for their flexibility, understanding, and encouragement, which greatly helped me balance my professional and academic commitments.

I am especially grateful to my friend Dr. Cristian Alfonso Gonzales Mora for his thoughtful and detailed feedback during the review process, which helped me significantly improve the quality of this thesis.

Finally, I acknowledge my supervisor, Daniel Matter, for his valuable guidance, constructive feedback, and continuous support during the development of this work. His insights have been instrumental in shaping both the technical and ethical dimensions of the thesis.

Abstract

The detection and classification of Nazi-related symbols in digital images are critical tasks in combating the spread of extremist content and supporting social science research on hate speech. This thesis investigates modern image classification methods for identifying Nazi symbols, including both binary classification (detecting whether any Nazi symbol is present) and multi-class classification (identifying the specific symbol type). A curated dataset was compiled, consisting of 99,435 non-Nazi images and 2,338 Nazi-related images spanning 22 symbol categories. Due to limited examples in some classes, several were regrouped into a consolidated *Neo-Nazi* category to improve model learnability.

Multiple Deep Learning (DL) architectures were evaluated, including You Only Look Once (YOLO), Residual Network (ResNet), EfficientNet, Vision Transformer (ViT), OpenCLIP, and Moondream. Additionally, classical machine learning models were explored using OpenCLIP visual embeddings as input features. Model performance was assessed using standard metrics such as accuracy, precision, recall, and F1-score.

Experimental results demonstrate that for the binary classification task, OpenCLIP embeddings combined with classical classifiers achieved competitive performance while requiring significantly fewer computational resources. However, in the multi-class classification task, this combination underperformed substantially compared to ViT and YOLO, with YOLO achieving the best overall performance in identifying specific symbol classes.

To support practical application, the best-performing models were integrated into an Application Programming Interface (API) service. A Continuous Integration/Continuous Delivery (CI/CD) pipeline was implemented to automate packaging and publishing of the service as a Docker image, enabling easy deployment by researchers or institutions. This work provides a lightweight and effective solution for classifying Nazi symbols in large-scale image datasets and contributes to broader efforts in content moderation and digital extremism research.

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1 Introduction

1.1 Background

In recent years, the internet—and particularly social media platforms—has become a key channel for the dissemination of extremist ideologies and hate-based content. This digital shift has transformed the landscape of radicalization, allowing fringe groups to amplify their messaging, coordinate activities, and propagate visual symbols of hate with unprecedented speed and reach. Among the most persistent and globally recognizable of these are Nazi symbols, which have been adopted and modified by neo-Nazi and far-right movements across different regions and time periods.

Unlike textual hate speech, which has been the subject of extensive moderation and analysis, harmful visual content remains comparatively under-researched and under-moderated. Symbols such as the swastika, SS bolts, Black Sun, and stylized versions of Adolf Hitler’s image are often embedded in memes, graffiti, protest photography, or propaganda posters—frequently in subtle, stylized, or degraded forms that evade both human and machine scrutiny. This creates a serious challenge for researchers, journalists, and civil society actors seeking to study or counteract online extremism.

From a technical standpoint, the classification of visual hate symbols represents a complex computer vision task. It lies at the intersection of iconographic recognition, fine-grained image classification, and open-set detection. The problem is further compounded by several practical constraints: the scarcity of annotated data, the subtle variation across symbol styles and contexts, and the risk of algorithmic bias or misuse in real-world deployments.

These challenges are not merely academic. Countries like Germany, Austria, and France have legal restrictions on Nazi imagery, and international organizations working in counter-extremism need scalable tools for monitoring hate symbols in open-source imagery. Despite this demand, there is a lack of open, well-documented, and computationally efficient systems capable of identifying these symbols across varied image sources and visual conditions. This gap motivates the development of a focused, ethically responsible classification framework—precisely the objective of this thesis.

1.2 Problem Statement

The automatic detection and classification of Nazi symbols in images is critical for researchers, journalists, civil society organizations, and online platforms working to monitor and mitigate the spread of extremist content. While commercial image classification services exist, they are often proprietary, lack transparency, and entail high usage costs—particularly for applications requiring large-scale image analysis or custom symbol detection. This presents a significant barrier not for governments per se, but for non-state actors and research groups who often operate under tight computational and financial constraints.

Even when national institutions have greater resources, scalability and cost-efficiency remain key considerations. High-throughput image classification tasks, such as analyzing large volumes of online content or social media images in near real-time, can demand substantial compute time and infrastructure. In practice, this limits the feasibility of using closed, black-box services that are either too slow, too expensive to scale, or unsuitable for customization. Moreover, reliance on opaque systems may hinder accountability and reproducibility—both essential in research and regulatory contexts.

To address these challenges, there is a need for an open, transparent, and efficient image classification system tailored specifically for Nazi symbol detection. Such a system should support both binary classification: detecting presence/absence of Nazi imagery, and multi-class classification: identifying specific symbols, while remaining lightweight and adaptable enough to run on accessible hardware like single-GPU workstations. The same requirements apply across adjacent domains, including online extremism tracking, digital forensics, and platform content moderation—making the solution broadly relevant beyond its immediate scope.

1.3 Objectives

This research aims to design, implement, and evaluate an efficient and robust image classification framework for the automatic detection and categorization of Nazi symbols in visual media. While the immediate focus is on Nazi iconography, the broader goal is to develop a generalizable and extensible approach that can be adapted to other categories of extremist or harmful visual symbols. This makes the framework applicable across domains such as civil society monitoring, journalism, and content moderation, where flexibility, interpretability, and scalability are critical.

The specific objectives of this study are as follows:

- **Dataset Construction:** Assemble a curated dataset of approximately 100,000 images, incorporating both Nazi-related and non-Nazi imagery to support supervised training, validation, and evaluation.
- **Annotation Strategy:** Structure and annotate the dataset to support both binary (presence/absence of Nazi symbols) and multi-class (identification of specific symbol classes)

classification tasks.

- **Model Benchmarking:** Implement and evaluate a diverse range of models for Nazi symbol classification, including Convolutional Neural Network (CNN)s (e.g., ResNet, EfficientNet), transformer-based models (e.g., ViT), and zero-shot vision-language models (e.g., OpenCLIP, Moondream). Repurpose YOLO as a classification model to leverage its real-time architecture. Explore hybrid approaches by combining OpenCLIP embeddings with classical machine learning classifiers (e.g., Support Vector Classifier (SVC), logistic regression, KNN) to benchmark performance across model families and representation strategies.
- **Performance Evaluation:** Assess model performance using standard classification metrics (e.g., accuracy, precision, recall, F1-score, and Area under the curve (AUC)), and conduct comparative analyses to identify strengths and limitations across model types.
- **Deployment Readiness:** Develop a lightweight and modular API service that enables efficient model inference via HTTP requests, facilitating integration into practical research and moderation workflows. The service is optimized for deployment on resource-constrained environments, such as single-GPU workstations.

Despite constraints in data coverage and computational resources, the proposed system—grounded in the above objectives—is designed to be modular, scalable, and adaptable. Its architecture allows for future extension to new symbol categories or broader visual moderation tasks through retraining, fine-tuning, or integration of new classifiers. This positions the framework as a practical and extensible foundation for real-world applications beyond the immediate use case of Nazi symbol classification.

1.4 Scope and Limitations

The visual symbols associated with Nazi and neo-Nazi ideologies are diverse, evolving, and context-dependent. Constructing a comprehensive dataset to cover all such variations is inherently challenging. This study therefore focuses on a representative subset of the most visually distinctive and widely recognized symbols. Specifically, the classification framework supports the following categories: *Black Sun*, *Flash and Circle*, *Broken Sun Cross*, *The Happy Merchant*, *Hitler*, *Nazi Salute*, *Atomwaffen Division Emblem*, *Blood & Honor Emblem*, *Celtic Cross*, *Combat 18 Emblem*, *Golden Dawn*, *Hammerskins Emblem*, *Identitarian Movement Emblem*, *Kolovrat*, *National Rebirth of Poland Emblem*, *Popular Front Emblem*, *Yellow Badge*, *Siegrune*, *Schutzstaffel (SS) Skull*, *Sturmabteilung Emblem*, *Swastika*, and *Wolfsangel*.

Two classification tasks are supported:

1. **Binary Classification:** Detecting the presence or absence of any Nazi-related symbol in a given image.
2. **Multi-Class Classification:** Identifying the specific symbol (among 13 target classes) when Nazi imagery is detected.

In addition to dataset limitations, there were also hardware constraints. All experiments were conducted on a single consumer-grade GPU (NVIDIA RTX 4060Ti, 16GB VRAM). This limited the ability to fine-tune large models or run memory-intensive experiments—particularly with transformer-based architectures (e.g., ViT) or multimodal embeddings (e.g., OpenCLIP). These models were used in this study, but only in lightweight or partially frozen configurations to stay within the available hardware limits. It also restricted the feasibility of training multiple models in parallel for ensembling. As a result, priority was given to computationally efficient architectures and lightweight training procedures that could fit within the available memory and runtime budget.

Despite these limitations, the system is designed to be modular and scalable. By updating the dataset and retraining or fine-tuning models, the framework can be extended to other extremist symbol categories or adapted to different contexts. The emphasis on transparency, efficiency, and accessibility makes the approach particularly suitable for researchers, journalists, and civil society organizations operating with limited infrastructure.

1.5 Thesis Structure

This thesis is organized as follows:

- **Chapter 2: Literature Review** – Surveys existing research on Nazi symbol detection and classification, emphasizing methodological trends, key findings, and identifying research gaps addressed in this work.
- **Chapter 3: Ethical Considerations** – Discusses the ethical implications of working with sensitive content, including concerns around data privacy, potential misuse, and broader responsibilities associated with automated hate symbol detection.
- **Chapter 4: Data and Model Development Methodology** – Describes the data collection process, dataset composition, preprocessing techniques, and the used classification algorithms. It also outlines the model training process, evaluation metrics, and justifies the chosen methodologies.
- **Chapter 5: Results** – Presents the experimental outcomes for binary and multi-class classification tasks, focusing on quantitative performance metrics (e.g., accuracy, precision,

recall, AUC). Detailed interpretation and comparative analysis are deferred to Chapter 6 (Discussion and Analysis).

- **Chapter 6: Discussion and Analysis** – Provides structured analysis and interpretation of experimental results in relation to the research objectives. This includes performance comparisons across model families, task-specific insights, discussion of high- and under-performing models, and reflections on system limitations, deployment considerations, and future improvements.
- **Chapter 7: Model Deployment and API Service** – Describes the deployment strategy for the best-performing models, including the architecture of the API service, the CI/CD pipeline, and details of the available endpoints.
- **Chapter 8: Conclusion and Future Work** – Summarizes the thesis contributions, reflects on key insights gained, and outlines directions for future research in the field of Nazi symbol classification.

2 Literature Review

This chapter reviews foundational and contemporary research relevant to the automated recognition of Nazi symbols in images. While direct studies on Nazi iconography remain limited due to ethical and legal sensitivities, substantial insights can be drawn from adjacent domains including visual symbol recognition [KSH12; Ian+15], hate content detection in social media [BA25; DTP19], multimodal meme analysis [Scr+21], and general-purpose image classification [Dos+21].

Particular emphasis is placed on methods for handling low-resource classification tasks, such as few-shot and zero-shot learning [Rad+21; Zha+24], which are highly relevant given the scarcity and variability of hate symbol imagery. The review also explores techniques for open-set recognition [Tüz+17]—a critical consideration in this work, where unseen or stylized symbol variants frequently appear in real-world data.

In addition, the chapter surveys research on dataset design and ethical considerations, including best practices for data transparency, annotation documentation, and the risks of deploying AI in politically sensitive contexts [Geb+21; SS17]. These studies help frame both the technical direction and ethical responsibility of this work, underscoring challenges such as visual ambiguity, adversarial content creation, and the importance of generalization across diverse symbol types.

Collectively, the literature lays a conceptual and methodological foundation for the design decisions made in this thesis—from model selection and training regimes to dataset structuring and evaluation strategies.

2.1 Visual Symbol and Logo Recognition

Foundational research in logo and symbol recognition has profoundly influenced methods for visual hate symbol detection. Iandola et al. [Ian+15] demonstrated the effectiveness of CNNs in recognizing logos within natural scenes—an analogous challenge to detecting Nazi symbols embedded in visually complex and cluttered environments such as posters, graffiti, or digital memes. Their work highlights the critical role of convolutional architectures in capturing localized visual patterns essential for precise symbol identification. Addressing scalability and intra-class variability, Wang et al. [Wan+19] introduced Logo-2K+, a large-scale dataset encompassing over 2,000 logo categories. This dataset emphasizes the importance of model generalization to distinguish visually similar but distinct classes, a challenge directly mirrored in the present work’s need to robustly differentiate various Nazi symbols from benign but visually related

imagery despite limited labeled data. Collectively, these foundational studies establish both the architectural framework and dataset design principles that inform the adaptation of convolutional models to the nuanced demands of hate symbol recognition.

Real-world deployment of Nazi symbol recognition systems must also accommodate significant visual variability—including stylistic modifications, degraded or occluded imagery, and previously unseen symbol variants. Tüzkö et al. [Tüz+17] tackled these challenges using open-set logo detection techniques that enable models to distinguish known categories from ambiguous or novel inputs. This open-set capability is crucial in practice, given that hate symbols frequently appear in altered or contextually obscured forms—whether partially hidden, stylized, or embedded within complex visual scenes such as memes or graffiti.

Guided by these insights, this study simulates open-set conditions through strategic dataset partitioning: *non-Nazi* images that are visually unrelated to the training data are included in the test set, while rare Nazi symbol categories are consolidated into a broader *neo-Nazi* class to promote generalization across scarce classes. Consequently, model evaluation extends beyond traditional closed-set accuracy metrics to incorporate robustness assessments under open-set-like conditions, thereby better reflecting real-world challenges and deployment scenarios.

Recent advances have explored weakly supervised approaches for symbol recognition [Tan+17], particularly suited to domains characterized by sparse or noisy annotations—common in extremist content datasets. Building on this, the current work leverages a pretrained visual encoder originally trained on large-scale general image datasets, which is then fine-tuned on a carefully curated hate symbol dataset annotated with domain-specific labels. This transfer learning strategy enables the model to retain broad visual recognition capabilities while adapting to the challenges of Nazi symbol classification, including limited data and high intra-class variation. While Tang et al. [Tan+17] focus on learning symbol localization from coarse image-level labels via saliency maps, the present approach leverages OpenCLIP’s vision-language alignment by comparing image embeddings with carefully designed textual prompts (e.g., “a swastika sign in the image”) to enable zero-shot classification based on semantic similarity. This reduces reliance on pixel-precise annotations while maintaining strong discriminative performance, aligning with the efficiency goals of weak supervision and avoiding the need for costly bounding box labels.

2.2 Hate Content Detection in Social Media and Memes

Hate symbol recognition increasingly intersects with multimodal content analysis, especially in the context of memes and online extremist propaganda. Belete and Alitasb [BA25] proposed a deep learning pipeline for detecting offensive Amharic-language memes by combining Bidirectional Long Short-Term Memory (BiLSTM) and dense networks, highlighting the effectiveness of jointly modeling textual and visual features. While their work is language-specific, it demonstrates a broader principle highly relevant to this research: Nazi symbols in contemporary

media often appear alongside provocative slogans or textual overlays, necessitating models that can parse both image and text. In alignment with this, the present work leverages OpenCLIP’s vision-language capabilities to perform image classification based on semantic similarity to textual prompts. This design choice enables symbolic content within complex meme compositions to be recognized without requiring localized annotations or object-level supervision.

Similarly, Das Bhattacharjee et al. [DTP19] employed a context-aware active learning framework to detect malicious online content, highlighting the importance of contextual cues. This is directly relevant to the current study, where hate symbols often appear in ambiguous visual contexts—such as being stylized, partially obscured, or embedded in complex imagery. To reflect these real-world challenges, the evaluation strategy incorporates open-set conditions by including *non-Nazi* test images that are visually dissimilar to the training data, as well as a *neo-Nazi* class formed from rare symbols. This setup tests the model’s ability to generalize beyond controlled training environments.

Furthermore, Scrivens et al. [Scr+21] analyzed how violent right-wing extremist groups use images to signal affiliation or ideology, often relying on covert symbolism. Their sociological findings reinforce the necessity of visual detection approaches that can handle subtle, disguised, or contextually loaded symbols. This perspective underpins the symbolic granularity of the classification schema in the current work, which goes beyond surface-level image matching to detect and disambiguate nuanced forms of Nazi symbolism encountered in user-generated online content.

2.3 Deep Learning Models for Visual Classification

Modern image classification systems largely rely on deep neural networks, which form the backbone of the symbol recognition pipeline developed in this study. AlexNet [KSH12] marked a pivotal advancement in DL by introducing a convolutional architecture capable of large-scale image classification. This laid the groundwork for subsequent developments in visual pattern recognition, which are essential when detecting visual hate symbols embedded in diverse and noisy real-world imagery.

VGGNet [SZ15] further demonstrated that deeper convolutional networks can effectively capture fine-grained visual details. This capability is particularly relevant to the challenges addressed here, as many Nazi symbols are small, partially obscured, or appear in degraded conditions—e.g., embedded within low-resolution memes or worn-out graffiti. While VGGNet itself is not directly used in this work, the core principle it highlights—that deep CNNs can learn hierarchical visual features—remains essential for symbol localization and class differentiation. This concept is reflected in the convolutional backbones adopted in this research, such as ResNet and EfficientNet, which build on the same representational strengths to support robust classification under such constraints.

More recently, ViT [Dos+21] have introduced a shift from localized convolutional operations to global self-attention mechanisms. This architectural innovation allows models to consider the entire image context when making predictions—an important advantage when Nazi symbols are entangled with background clutter or positioned alongside provocative text. In this study, ViT serves as one of the baseline architectures to benchmark performance in such visually complex scenarios, complementing convolution-based models and offering comparative insights into context-aware symbol classification.

By evaluating both CNN-based and Transformer-based models, this research identifies the architectures most effective at handling the variability, subtlety, and complexity inherent in extremist visual content.

2.4 Zero-Shot and Few-Shot Symbol Recognition

A persistent challenge in Nazi symbol classification lies in the scarcity of labeled training data, particularly for rare, emerging, or region-specific variations. These limitations make it difficult to apply fully supervised methods, motivating the adoption of zero-shot and few-shot learning strategies to support generalization in low-resource settings.

Xian et al. [Xia+19] conducted a comprehensive benchmark of Zero-Shot Learning (ZSL) approaches, emphasizing the trade-offs between generalization and domain adaptation—an issue especially relevant when trying to recognize visually diverse hate symbols with limited examples. Their findings inform the current approach, which relies on pretrained vision-language models to compensate for sparse supervision.

CLIP, introduced by Radford et al. [Rad+21], aligns image and text representations through contrastive learning [Che+20], enabling classifiers to associate symbolic imagery with descriptive prompts even without class-specific training. This capability is leveraged via OpenCLIP embeddings, allowing Nazi-related symbols to be recognized based on semantic cues such as “swastika” or “SS emblem,” thus facilitating zero-shot classification for underrepresented or novel categories.

DiffCLIP [Zha+24] extends this paradigm by enabling prompt tuning for few-shot scenarios. Building on this idea, the research employs carefully engineered prompts and domain-curated symbol descriptions to adapt the pretrained model to hate symbol classification without requiring extensive per-class supervision. This is particularly beneficial when dealing with minority classes, such as lesser-known neo-Nazi symbols, or images exhibiting stylistic alterations.

Relation Network (RN) [Sun+18] offer another complementary perspective by classifying unseen categories based on similarity in feature space. While RNs are not directly implemented, the underlying idea of distance-based matching informs the use of classic machine learning models (e.g., k-NN, SVM) operating on CLIP-derived embeddings. This design choice enables the system to generalize beyond the training distribution—capturing subtle visual relationships

and supporting recognition of stylized or obscured symbols not seen during training.

Together, these zero- and few-shot learning frameworks provide essential scaffolding for symbol recognition in settings where labeled data are scarce, guiding the architecture and training strategies used in this research to address both generalization and semantic alignment.

2.5 Dataset Transparency and Ethical Considerations

The sensitive and potentially harmful nature of Nazi symbol detection demands rigorous attention to transparency, accountability, and ethical responsibility throughout the machine learning lifecycle. Gebru et al. [Geb+21] introduced the “datasheets for datasets” framework, advocating for structured documentation that details dataset provenance, collection methods, annotation processes, and potential biases. This approach is directly applied in the present research through detailed dataset reporting, including symbol class distributions, labeling rationale, and preprocessing decisions. By adopting such practices, the project ensures that downstream users and evaluators can understand the dataset’s limitations, sources, and intended use cases—crucial when handling politically charged or culturally sensitive content.

In parallel, Shahriari [SS17] highlights the importance of ethically aligned AI design, emphasizing principles such as human rights, safety, transparency, and fairness. These considerations directly inform the system’s development, especially given the risks associated with false positives (e.g., benign religious or cultural symbols misclassified as hate-related) or unintended surveillance use. In response, the research includes deliberate measures to reduce harm: symbol definitions are carefully grounded in recognized legal and sociological classifications, performance metrics are stratified to monitor class-wise precision and recall, and semantic alignment through prompt-based classification supports limited interpretability by linking predictions to human-readable concepts.

Together, these ethical frameworks shape not only how the dataset is constructed and documented, but also how the classifier is evaluated, interpreted, and contextualized. This commitment to transparency and responsible design aligns the technical contributions with the broader societal impact of deploying hate symbol recognition systems in practice.

3 Ethical Considerations

The development of ML systems for the classification of Nazi symbols entails a range of ethical challenges due to the sensitive, harmful, and potentially illegal nature of the target content. These concerns reflect broader societal expectations for responsible AI, as articulated in international frameworks such as United Nations Educational, Scientific and Cultural Organization (UNESCO)’s *Recommendation on the Ethics of Artificial Intelligence* [UNE21]. This chapter outlines the core ethical principles guiding the research, grounded in widely recognized frameworks such as the Institute of Electrical and Electronics Engineers (IEEE) *Ethically Aligned Design* guidelines [SS17], the EU AI Act [Com23], and the Deutsche Forschungsgemeinschaft (DFG)’s *Guidelines for Safeguarding Good Research Practice* [Deu]. These considerations are essential for ensuring that the study is conducted responsibly, complies with legal norms, and anticipates the social implications of deploying such a system.

- **Foundational Ethical Frameworks:** This research is informed by established ethical guidelines such as the IEEE Ethically Aligned Design [SS17] and the Belmont Report [PR79], the latter—originally developed for biomedical research—offers enduring principles like respect for persons and beneficence that extend to socially impactful AI systems. These principles guide our handling of sensitive data and system design choices.
- **Handling of Sensitive Content:** Nazi symbols are strongly associated with hate, violence, and genocide. Datasets containing such imagery must be handled with care to avoid trivializing or normalizing hateful ideologies. Access to these datasets is strictly limited, and appropriate content warnings and disclaimers are used. Data management follows established protocols for working with sensitive or controversial materials, in accordance with research integrity standards outlined by the DFG [Deu].
- **Purpose and Intent:** The goal of this research is to assist in identifying and moderating harmful content, not to promote or propagate extremist ideologies. Transparency in the project’s objectives helps prevent misuse, aligning with principles of socially beneficial AI outlined by the IEEE and UNESCO [SS17; UNE21].
- **Legal Compliance:** In countries such as Germany and Austria, strict laws (e.g., Section 86a of the German Criminal Code) prohibit the display and distribution of Nazi symbols. This study complies with all applicable regulations, ensuring that content is accessed solely for legitimate research purposes and not redistributed [23].

- **Data Sourcing and Annotation:** All datasets were collected from publicly available repositories—primarily Kaggle and Roboflow—which permit academic and non-commercial use under their respective licenses¹. No restricted or proprietary content was used. All annotations were conducted manually by the researcher. Content warnings were observed throughout the process to mitigate potential psychological distress, as also described in Section 4.1 (Dataset Construction and Symbol Taxonomy) from Chapter 4 (Data and Model Development Methodology).
- **Bias and Misclassification Risks:** Classification systems may mistakenly flag benign content or overlook altered forms of hate symbols. Such errors could lead to unjust moderation or missed threats. To address this, the study prioritized balanced training data, fairness-aware evaluation, and extensive model validation in line with AI fairness principles [Meh+22].
- **Dual-Use Concerns:** While developed for moderation purposes, such a classification system could be repurposed for malicious use, such as evading content filters or enabling authoritarian surveillance. Recognizing this dual-use risk [Bru+24], the research limits model distribution and recommends transparency, access control, and red-teaming before deployment.
- **Privacy and Consent:** If any images included identifiable individuals, anonymization procedures were applied in accordance with General Data Protection Regulation (GDPR) principles and ethical AI guidelines [16]. No personally identifiable information (PII) was intentionally collected or retained.
- **Transparency and Accountability:** All stages of the research—including data provenance, preprocessing, model selection, and limitations—are documented and disclosed. This transparency aligns with reproducible research norms and promotes accountability in the deployment of AI systems on sensitive content [Mit+19].

In sum, these ethical frameworks are not peripheral to the study but foundational to its design and execution. They inform not only how data is sourced, labeled, and secured, but also how models are evaluated, interpreted, and constrained in their use. With these considerations in place, the following chapter details the methodological pipeline through which the classification system is developed and validated.

¹Kaggle Terms of Service: <https://www.kaggle.com/terms>; Roboflow Terms: <https://roboflow.com/terms>

4 Data and Model Development Methodology

This chapter outlines the methodology adopted for developing and evaluating the proposed Nazi symbol classification system. It begins with a detailed overview of the dataset construction process, including the sources of imagery and structuring strategies tailored for both binary and multi-class classification tasks. The chapter then describes the taxonomy of Nazi symbols included in the dataset, offering historical and visual context for each symbol class. A summary of the distribution of these symbols is provided in Table 4.2, while the relative proportions of the binary and multi-class datasets are visualized in Figure 4.3 and Figure 4.4, respectively. Subsequently, the ML and DL models used are introduced, followed by the preprocessing pipeline, training procedures, and hardware configurations. Finally, the evaluation metrics used to assess model performance are presented, forming the foundation for the experimental results in the subsequent chapter.

4.1 Dataset Construction and Symbol Taxonomy

To develop a reliable image classification system capable of identifying Nazi symbols in diverse visual contexts, a custom dataset was curated from multiple public sources. This dataset supports both binary and multi-class classification tasks by incorporating a broad range of Nazi and non-Nazi imagery, annotated and structured with attention to historical accuracy, visual clarity, and class balance.

4.1.1 Data Sources

The dataset was assembled from two primary sources:

- **Roboflow**¹: Provided labeled datasets containing images of known Nazi symbols such as the swastika, SS bolts, and Black Sun. These were selected for their relevance, annotation quality, and visual variability, forming the positive class.
- **Kaggle**²: Contributed general-purpose datasets consisting of images unrelated to extremism, including scenes from daily life, commercial logos, and various public domains. These

¹<https://roboflow.com>

²<https://www.kaggle.com>

constituted the negative class, essential for training models to distinguish hate symbols from benign visual content.

All datasets were obtained under open-use or research-oriented licenses, with careful review of usage terms and ethical considerations.

4.1.2 Preprocessing and Quality Control

To ensure dataset reliability and suitability for training, the following preprocessing steps were applied:

- **Manual Verification:** All symbol annotations were manually reviewed to verify label correctness and remove noise.
- **Symbol Taxonomy Alignment:** Nazi symbols were categorized based on visual features and historical context, referencing established taxonomies from the Anti-Defamation League (ADL) [13] and academic literature on extremist symbology.
- **Deduplication:** Duplicate and near-duplicate images were removed to reduce overfitting and prevent data leakage.
- **Partitioning:** The dataset was randomly split into 70% training, 15% validation, and 15% test sets, ensuring no overlap of symbol instances across splits.

A summary of all datasets is provided in Table 4.1, and representative examples from the positive and negative classes are shown in Figure 4.1.

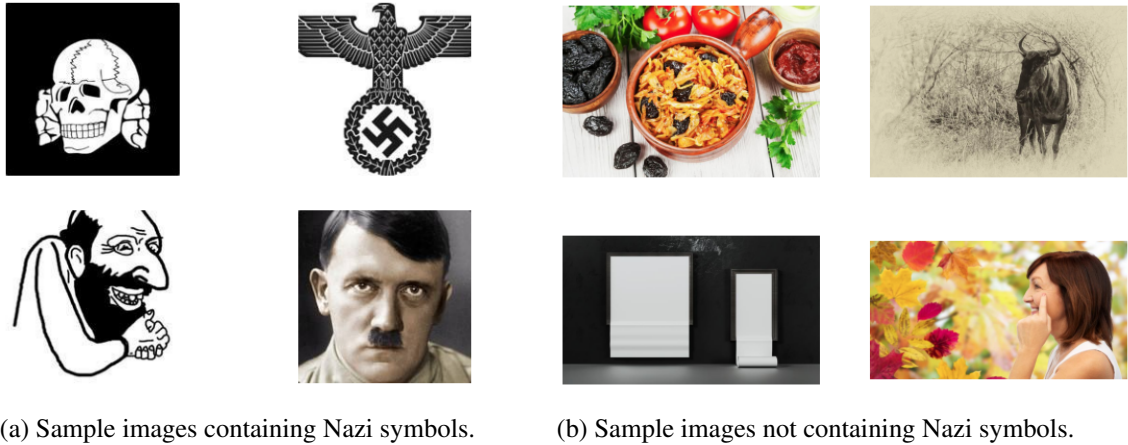


Figure 4.1: Example of positive and negative classes used for training.

Platform	Dataset Name / Description	Purpose	Notes
Roboflow	detect Dataset [mar23]	Positive Class	Swastika and related Nazi symbols
Roboflow	swastika Dataset (imgann) [img23a]	Positive Class	Swastika symbol detection
Roboflow	swastika Dataset (Darja Nechaeva) [Nec22]	Positive Class	Historical swastika variations
Roboflow	swastika_detect Dataset [Inc23]	Positive Class	Swastika detection for moderation
Roboflow	symbols Dataset [Vol23]	Positive Class	Includes general and hate symbols
Roboflow	violent object Dataset [img23b]	Positive Class	Hate symbols and violent imagery
Roboflow	Swastiki_Net Dataset [He624]	Positive Class	Swastika symbol classification
Roboflow	Content Moderation Dataset [Tec24]	Positive Class	Multiple harmful or extremist symbols
Roboflow	Swastika_Detection Dataset [Kav22]	Positive Class	Swastikas in different contexts
Kaggle	100 Sports Image Classification [Ger23]	Negative Class	Neutral sports images
Kaggle	Car vs Bike Classification Dataset [Dee22]	Negative Class	Background class diversity
Kaggle	People Clothing Segmentation [Lak21]	Negative Class	Background objects, neutral content
Kaggle	Fruits and Vegetables Dataset [SET22]	Negative Class	For reducing false positives
Kaggle	Animal Image Dataset [BAN22]	Negative Class	Animal images for neutral comparison
Kaggle	Kids and Adults Detection [GIH21]	Negative Class	Human figures, non-extremist
Kaggle	SCOOBY DOO Classification [DEL25]	Negative Class	Cartoon data for variance
Kaggle	Low Light Animals [Sar25]	Negative Class	Low-light conditions for robustness
Kaggle	AI vs Human-Generated Images [Shu25]	Negative Class	Diverse image source simulation

Table 4.1: Summary of datasets used in this study for training, validation, and testing.

4.1.3 Symbol Taxonomy and Description

The collected dataset includes 22 distinct symbols associated with Nazi ideology, grouped based on their historical origin, contemporary usage, and visual form. These symbols were selected due to their recognized use in Nazi Germany and/or modern neo-Nazi and white supremacist movements. Figure 4.2 provides a visual overview.

The following list is a concise description of each symbol class included in the dataset:

- **Nazi Salute:** The Sieg Heil gesture used to demonstrate allegiance to Hitler; currently banned in several countries.
- **Flash and Circle:** A lightning bolt within a circle, originally used by the British Union of Fascists; later adopted by various white supremacist groups.
- **Sturmabteilung (SA) Emblem:** Emblem of the Nazi Party’s original paramilitary wing, representing early violent enforcement of Nazi ideology.
- **Combat 18 Emblem:** Insignia of a violent neo-Nazi group, with “18” representing Adolf Hitler by numerical substitution.
- **SS Skull (Totenkopf):** A skull symbol signifying loyalty unto death, worn by SS personnel, especially in concentration camps.

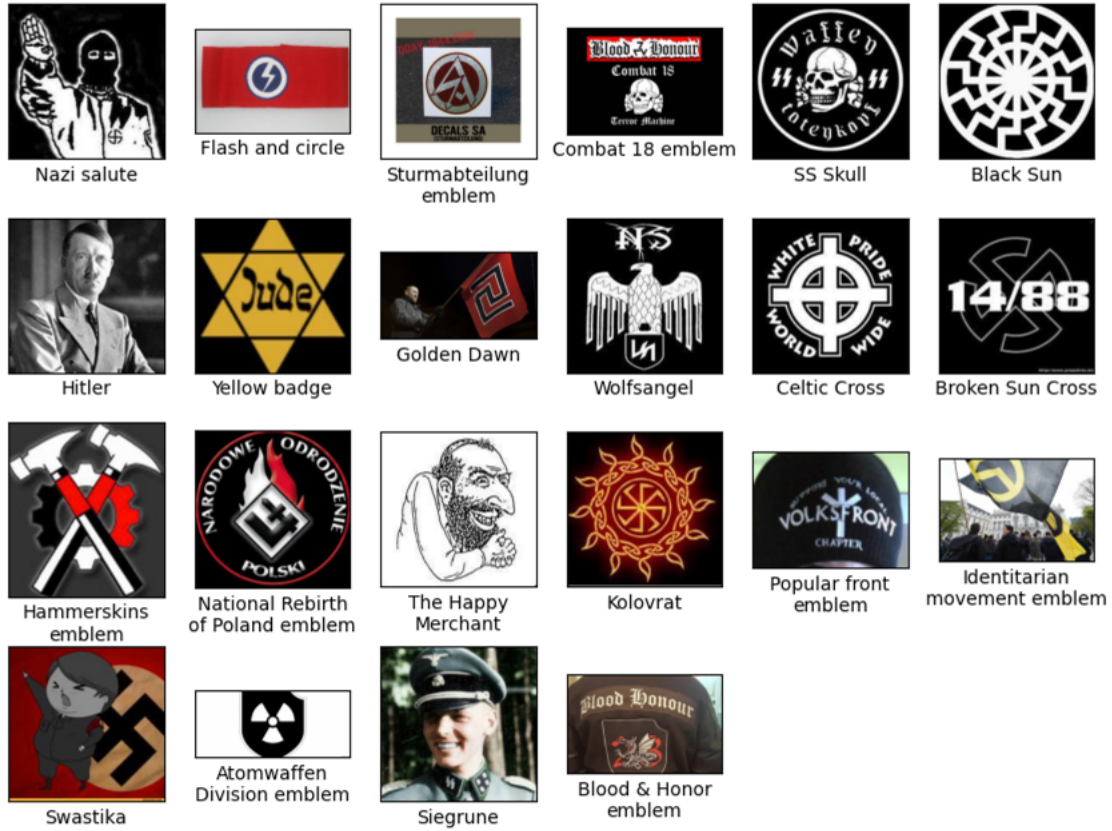


Figure 4.2: Example images from 22 Nazi symbol classes.

- **Black Sun:** A circular symbol composed of twelve radial Sig runes, associated with SS mysticism and adopted by neo-Nazi groups as an esoteric representation of Aryan heritage.
- **Hitler:** Images of Adolf Hitler, representing the central figure of Nazi ideology.
- **Yellow Badge:** A yellow Star of David used to identify Jews during the Holocaust, symbolizing systemic persecution.
- **Golden Dawn:** Emblem of a Greek far-right political party with known neo-Nazi affiliations and violent activity.
- **Wolfsangel:** Originally a medieval glyph, later adopted by Nazi divisions and modern neo-Nazi movements.
- **Celtic Cross:** While historically religious, its simplified form is now a common white nationalist symbol.

- **Broken Sun Cross:** A variation of the ancient sun cross with swastika-like arms; historically linked to Nazi-affiliated movements and Waffen-SS units.
- **Hammerskins Emblem:** Two crossed hammers representing a white supremacist skinhead group involved in hate music and propaganda.
- **National Rebirth of Poland (NOP):** Symbol of a far-right Polish nationalist group promoting xenophobic and anti-Semitic ideologies.
- **Happy Merchant:** A racist caricature of a Jewish man, originating from hate propaganda and widely circulated in white supremacist memes.
- **Kolovrat:** An ancient Slavic sun symbol re-appropriated by white supremacist groups in Eastern Europe.
- **Popular Front Emblem:** Used by fascist-aligned organizations; typically combines nationalist imagery with militaristic symbols.
- **Identitarian Movement Emblem:** A yellow circle with a black Greek lambda, symbolizing European ethno-nationalism and anti-immigration rhetoric.
- **Swastika:** A symbol appropriated by the Nazis to signify racial purity and nationalism; now the most recognizable icon of hate worldwide.
- **Atomwaffen Division Emblem:** A stylized nuclear symbol used by a neo-Nazi terrorist group advocating for white supremacist accelerationism.
- **Siegrune:** An SS lightning-bolt rune adapted from Germanic runes, widely used by the Nazi Schutzstaffel.
- **Blood and Honour Emblem:** Derived from the Hitler Youth motto “Blut und Ehre”; used by an international neo-Nazi network promoting white nationalist music.

4.1.4 Task-Oriented Dataset Structuring

To simulate open-set conditions and account for class imbalance, the dataset was structured with deliberate design choices. Several Nazi-related categories with limited samples were grouped into a new composite class labeled *neo-Nazi*, allowing the model to generalize over symbol variants while mitigating data sparsity. Additionally, for the binary classification task, *non-Nazi* images were randomly split into training, validation, and test sets. As a result, many *non-Nazi* samples in the validation and test sets had no visual or semantic overlap with training examples. This setup approximates real-world open-set scenarios, where models must detect unfamiliar or previously unseen content.

The dataset was structured to serve two complementary classification tasks as follows:

- **Binary Classification (Symbol Detection):** All images containing any of the 22 Nazi-related symbols were labeled as *Nazi*, while the remaining images were labeled *non-Nazi*. The dataset includes approximately 2,300 positive and 99,000 negative images, resulting in a class ratio of approximately 1:42.
- **Multi-Class Classification (Symbol Identification):** Only images from the 22 Nazi symbol categories were included. Each retained its class-specific label, totaling around 2,300 images. Due to severe class imbalance, symbol classes with very few samples were grouped into a new composite category labeled *neo-Nazi* (see Table 4.2), improving balance and model generalizability without sacrificing interpretability.

Class	Images
Black Sun	152
Flash and circle	13
Broken Sun Cross	21
The Happy Merchant	39
Hitler	228
Nazi salute	6
neo-Nazi	188
Yellow badge	4
Siegrune	311
SS Skull	257
Sturmabteilung emblem	11
Swastika	1070
Wolfsangel	38

(a) Regrouped symbol distribution

Class	Images
Atomwaffen Division	8
Blood & Honor emblem	25
Celtic Cross	51
Combat 18 emblem	6
Golden Dawn	19
Hammerskins emblem	12
Identitarian movement	11
Kolovrat	38
National Rebirth of Poland	10
Popular Front emblem	8

(b) neo-Nazi Symbol distribution

Table 4.2: Symbol distribution

All dataset collection and usage procedures adhered to recognized ethical research standards, including the IEEE Ethically Aligned Design guidelines [SS17] and the Belmont Report [PR79]. No personally identifiable information was included, and symbol use was limited to academic research under controlled and justified purposes, as further detailed in Chapter 3.

4.2 Classification Models

To rigorously assess the efficacy of various ML methodologies in the detection and classification of Nazi symbols, experiments using a diverse set of models were conducted. This collection

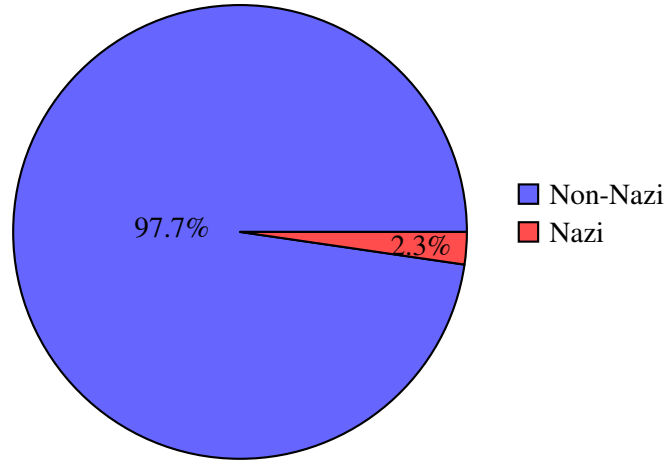


Figure 4.3: Pie chart showing binary classification dataset distribution.

comprises both state of the art (SotA) DL architectures and traditional ML algorithms, enabling a comprehensive comparison across different computational paradigms and levels of model complexity.

4.2.1 Real-Time Object Detection (You Only Look Once: YOLO)

YOLO is a real-time object detection framework that formulates detection as a unified regression problem, simultaneously predicting bounding boxes and class probabilities directly from full images. By processing the entire image in a single forward pass, YOLO achieves a favorable trade-off between inference speed and accuracy, rendering it highly suitable for scenarios demanding real-time processing. Although primarily designed for object detection, YOLO can be adapted for classification tasks by assigning the class label corresponding to the highest-confidence detection. In this study, the pretrained `YOLOv11-cls` classification model was utilized, which benefits from improved backbone architectures and refined label assignment strategies that enhance both robustness and computational efficiency.

4.2.2 Residual Network (ResNet)

ResNet is a CNN architecture that introduces residual learning through identity shortcut connections, thereby mitigating the vanishing gradient problem often encountered in training deep networks [He+15]. This design facilitates the effective training of substantially deep architectures. The ResNet family has consistently demonstrated superior performance in image classification benchmarks. There was specifically employed `ResNet34`, which offers a well-established balance between depth, computational cost, and classification accuracy.

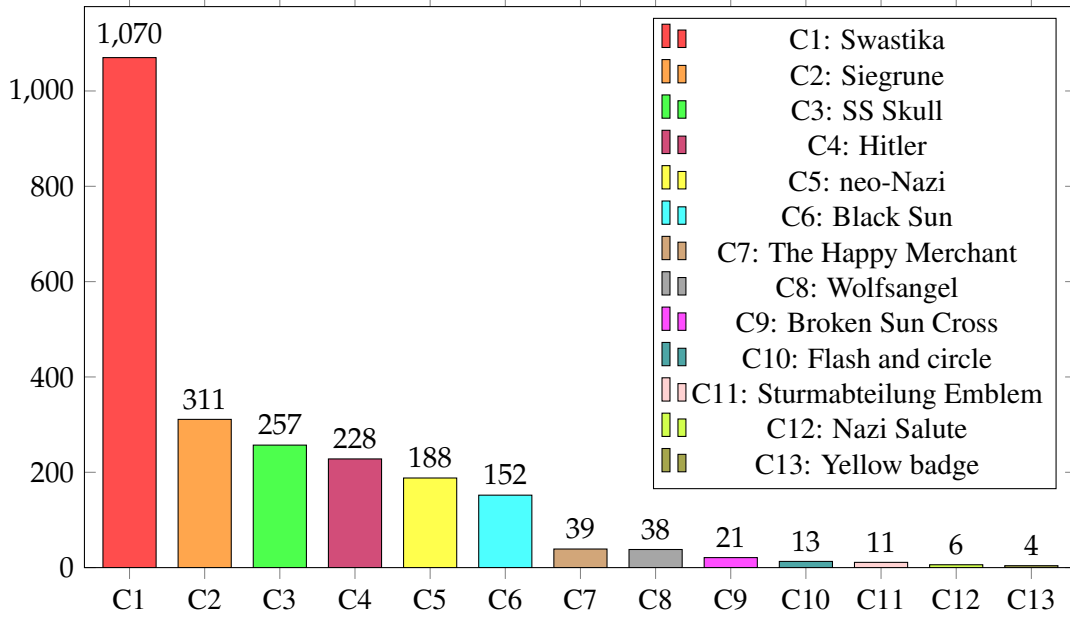


Figure 4.4: Dataset distribution of the multi-class classification task.

The ResNet34 model used in this study is pretrained on the ImageNet dataset, enabling effective transfer learning for the target classification tasks.

4.2.3 EfficientNet

EfficientNet represents a family of CNNs that leverage a compound scaling method to uniformly scale network depth, width, and resolution. This principled approach yields models that achieve SotA accuracy with significantly fewer parameters and computational resources. Due to its efficiency, EfficientNet is particularly well-suited for large-scale image classification tasks under constrained hardware conditions. In this thesis, the EfficientNet-B2 variant was employed, which offers a favorable trade-off between model complexity and accuracy. It serves as a lightweight yet competitive alternative to deeper, more computationally intensive networks in our experiments. The EfficientNet-B2 model used is pretrained on the ImageNet dataset, enabling transfer learning to improve performance on the target task.

4.2.4 Vision Transformer (ViT)

The ViT model innovatively adapts the transformer architecture, originally devised for Natural Language Processing (NLP), to vision tasks by segmenting images into fixed-size patches, which are then treated as a sequence of tokens. It avoids traditional convolutional operations entirely,

instead relying on self-attention mechanisms to model long-range dependencies and capture global contextual information. ViT has demonstrated performance on par with or exceeding that of conventional CNNs, particularly when trained on large datasets.

This study employs the pretrained `vit_base_patch16_224` model to evaluate the effectiveness of transformer-based architectures in classifying visual symbols that exhibit complex semantic patterns.

4.2.5 OpenCLIP

OpenCLIP is an open-source reimplementation of Contrastive Language-Image Pre-training (CLIP), a multimodal vision-language model trained on extensive datasets of image-text pairs. By learning a shared embedding space for both modalities, OpenCLIP enables zero-shot classification through similarity comparison between image embeddings and text prompt embeddings. In this study, OpenCLIP is employed in two distinct modes: zero-shot inference and as a fixed feature extractor, providing a flexible and semantically enriched representation aligned with natural language descriptions.

4.2.6 Moondream

Moondream is a compact multimodal model optimized for low-resource vision-language tasks. Although not explicitly designed for image classification, it can be adapted for lightweight classification via prompt-based image-text similarity comparisons. Its strength lies in efficient on-device inference, making it particularly advantageous for deployment in computationally constrained environments such as mobile or embedded systems. Moondream was investigated as a potential candidate for these resource-limited classification scenarios.

4.2.7 OpenCLIP as Encoder with Classical ML Classifiers

To harness the semantic richness of OpenCLIP embeddings, OpenCLIP was employed as a frozen feature encoder, generating fixed-length vector representations for input images. These embeddings subsequently serve as input to a variety of traditional ML classifiers. This two-stage pipeline facilitates rapid experimentation with interpretable, computationally efficient models, especially beneficial for small datasets or when DL approaches incur excessive computational costs. A detailed summary of the key characteristics of the classification algorithms employed is presented in Table 4.3.

Classic Classifiers Used:

- **Logistic Regression (LR):** A linear probabilistic model well-suited for binary classification tasks.

- **Stochastic Gradient Descent (SGD) Classifier:** An efficient linear classifier trained using stochastic gradient descent, scalable to large datasets.
- **KNeighbors Classifier:** A classifier implementation of K-Nearest Neighbors (KNN), which is a non-parametric, instance-based learning algorithm that assigns labels based on the proximity of training examples in feature space.
- **SVC:** A margin-based classifier effective in high-dimensional feature spaces, leveraging kernel methods to handle non-linear separability.
- **Decision Tree Classifier:** A hierarchical model that partitions the feature space via threshold-based splits, enabling interpretable decision rules.
- **Bagging Classifier:** An ensemble approach that aggregates multiple decision trees trained on bootstrapped samples to reduce variance and improve stability.
- **Ada Boost Classifier:** A sequential ensemble method that iteratively focuses on misclassified samples to enhance overall predictive accuracy.
- **Gradient Boosting Classifier:** Constructs strong predictive models through stage-wise optimization of a differentiable loss function using weak learners.
- **Random Forest Classifier:** An ensemble of decision trees trained with random feature selection, providing robustness against overfitting and improved generalization.
- **Voting Classifier:** Combines the predictions of multiple base models (e.g., LR, Random Forest, SVC) using soft or hard voting strategies to enhance classification performance.

Classifier	Model Type	Linear	Ensemble	Interpretable	Scalable
Logistic Regression	Parametric	Yes	No	High	High
SGD Classifier	Parametric	Yes	No	Moderate	Very High
KNeighbors Classifier	Instance-based	No	No	Low	Low
SVC	Parametric	Yes	No	Moderate	High
Decision Tree Classifier	Non-parametric	No	No	High	Moderate
Bagging Classifier	Non-parametric	No	Yes	Moderate	High
AdaBoost Classifier	Non-parametric	No	Yes	Low to Moderate	Moderate
Gradient Boosting Classifier	Non-parametric	No	Yes	Low	Moderate
Random Forest Classifier	Non-parametric	No	Yes	Moderate	High
Voting Classifier	Hybrid	Depends on estimators	Yes	Low	High

Table 4.3: Summary of classification algorithms and their key characteristics

4.2.8 Notation for Classification Models

For brevity and consistency, the following shorthand notations will be used throughout the remainder of this thesis:

- **YOLO** refers to the YOLOv11 model.
- **ResNet** refers to ResNet34.
- **EfficientNet** refers to EfficientNet-B2.
- **ViT** refers to vit_base_patch16_224.

These notations are adopted solely for readability and convenience within the scope of this thesis and do not imply reference to the entire family of model architectures.

4.3 Data Preprocessing

To ensure the reliability and consistency of model training, a comprehensive data preprocessing pipeline comprising resizing, normalization, and offline data augmentation was implemented. These preprocessing steps are critical to mitigating noise, simulating real-world variations, and facilitating efficient model convergence.

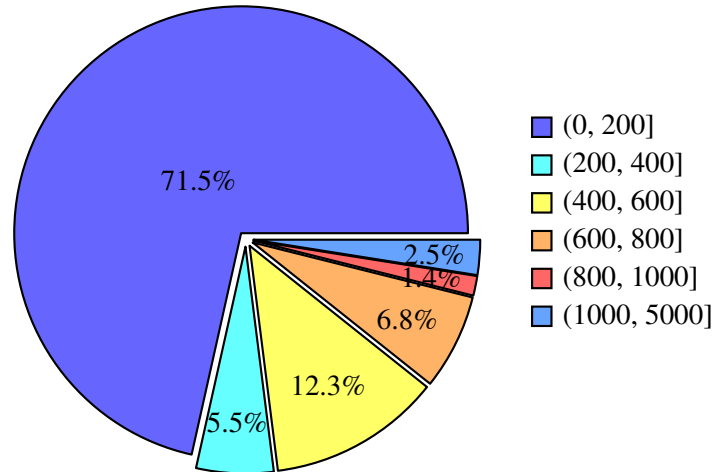


Figure 4.5: Distribution of image sizes in the dataset, grouped by pixel width and height. The majority of images fall into the (0, 200] range.

- **Resizing:** All images were resized to a uniform resolution of 224×224 pixels. This choice was driven by two main factors. First, as illustrated in Figure 4.5, analysis of the dataset revealed that approximately 71% of Nazi symbol images possess native dimensions smaller than 200×200 pixels, while only 5.5% falls within the 200×200 to 400×400 range. Standardizing to 224×224 pixels minimizes distortion and loss of detail for the majority of images. Second, several evaluated architectures, including `vit_base_patch16_224`, require fixed input sizes due to their structural constraints. Adopting this resolution facilitates compatibility across models and ensures consistent comparative evaluation.
- **Normalization:** Pixel intensities were scaled to the range $[0, 1]$ on a per-image basis. This normalization stabilizes the optimization process and expedites convergence by ensuring consistent data scaling.
- **Data Augmentation:** Offline data augmentation was employed to artificially enhance the diversity of the training set and improve model generalization. Augmentations were applied probabilistically to each sample to simulate realistic variations encountered in operational contexts. The augmentation techniques included:
 - Conversion to grayscale, reducing reliance on color information.
 - Automatic contrast adjustment, normalizing image dynamic range.
 - Random horizontal and vertical flipping with a 50% probability, simulating orientation variability.
 - Random rotations within $\pm 15^\circ$, introducing viewpoint variability.
 - Random shearing (vertically and horizontally within $\pm 10^\circ$), simulating perspective distortions.
 - Random adjustments to hue ($\pm 15^\circ$), saturation (± 63 units), and brightness (± 38 units), increasing robustness to lighting and color shifts.
 - Random exposure modifications to simulate overexposure and underexposure conditions.
 - Gaussian blur applied randomly to emulate realistic image imperfections.
 - Sparse addition of random noise (0.1% probability) to enhance resilience against sensor noise and degradation.

These augmentations reflect common distortions prevalent in real-world imagery, such as variable lighting, camera movement, compression artifacts, and historical degradation, which are particularly pertinent for Nazi symbol detection tasks.

All preprocessing operations were implemented using a combination of `Pillow` and `OpenCV` libraries during an offline stage. The processing pipeline applies resizing first, followed by

augmentation and normalization, thereby ensuring computational efficiency and uniformity across all samples.

Examples illustrating the data preprocessing steps—comprising augmentation, resizing, and normalization—are depicted in Figure 4.6.



(a) Original image example.



(b) Augmented image example.



(c) Resized image example.



(d) Normalized image example.

Figure 4.6: Illustrative examples of data preprocessing steps.

4.4 Model Training

Each model employed in this study was trained with hyperparameter configurations carefully tailored to its architecture and the specific classification task at hand. To mitigate overfitting, early stopping was implemented for all DL models, monitoring validation loss with a patience of 10 epochs. The model checkpoint achieving the best validation performance was preserved for subsequent evaluation. A summary of the primary training hyperparameters is provided in Table 4.4.

ResNet was trained for 100 epochs using a batch size of 48 and an input resolution of 224×224 pixels. The binary classification tasks employed the Binary cross-entropy (BCE) loss, whereas multi-class problems utilized the categorical cross-entropy (CCE) loss function. Optimization

was performed using the Adam algorithm with an initial learning rate of 1×10^{-3} and weight decay set to 1×10^{-5} . A 1-cycle learning rate scheduler was adopted to dynamically adjust the learning rate throughout training.

EfficientNet training spanned 100 epochs with a batch size of 48 and input size 224×224 . For binary classification, the Nesterov momentum variant of Adam (NAdam) optimizer was used with a learning rate of 2×10^{-3} , no weight decay, and momentum decay of 4×10^{-3} . For multi-class tasks, Adam optimizer was applied with a learning rate of 1×10^{-3} . The model’s fully connected classification head was replaced with a linear layer matching the number of target classes.

ViT was trained for 100 epochs with batch size 48 and input resolution 224×224 . The Adam optimizer with a learning rate of 2×10^{-5} was utilized. Loss functions were selected based on the task: BCE for binary classification and CCE for multi-class scenarios.

YOLO underwent training for 100 epochs, batch size 48, and input size 224×224 . The SGD optimizer was employed with an initial learning rate of 0.01, momentum 0.9, and weight decay 5×10^{-4} . Correspondingly, BCE and CCE losses were applied for binary and multi-class classification, respectively.

OpenCLIP-based models leveraged OpenCLIP as a fixed feature extractor to obtain embeddings from resized input images (224×224). Subsequently, classical ML classifiers such as LR, SVC, and KNeighborsClassifier were trained on these embeddings using CPU resources.

Zero-shot classification approaches with OpenCLIP and Moondream circumvented dataset-specific training entirely; classification was performed by computing cosine similarity between image and text embeddings, enabling direct inference without fine-tuning.

All training procedures were executed on an NVIDIA RTX 4060Ti GPU with 16 GB of VRAM.

Learning rates were not unified across all models. Instead, they were selected based on architecture-specific best practices and empirical tuning. Transformer-based models such as ViT are sensitive to large gradient steps and thus require lower learning rates for stable convergence. Conversely, convolutional architectures like ResNet and EfficientNet can tolerate moderately higher learning rates. The values chosen reflect typical recommendations from the literature and were validated to ensure consistent training dynamics.

4.4.1 Prompt Design for Zero-Shot Classification

For zero-shot classification with OpenCLIP and Moondream, textual prompts were carefully designed to capture the distinctive visual and historical features of each Nazi symbol class. These prompts enable the models to relate image content to semantic concepts via image-text similarity

Model	Epochs	Batch Size	Input Size	Loss Function	Optimizer	Learning Rate
ResNet-34	100	48	224×224	BCE/CCE ¹	Adam	1e-3
EfficientNet-b2 (binary)	100	48	224×224	BCE	Nadam	2e-3
EfficientNet-b2 (multi)	100	48	224×224	CCE	Adam	1e-3
ViT	100	48	224×224	BCE/CCE	Adam	2e-5
YOLOv11	100	48	224×224	BCE/CCE	SGD	0.01
OpenCLIP + ML	–	–	224×224	–	–	–
Zero-shot (OpenCLIP, Moondream) ²	–	–	–	–	–	–

¹ BCE = Binary Cross-Entropy, CCE = Categorical Cross-Entropy.

² Zero-shot models are not trained on the dataset; they use pretrained weights and inference directly.

Table 4.4: Summary of training hyperparameters used for each model.

scores or multimodal reasoning. Moondream’s prompts take the form of analytical questions, eliciting binary (“Yes”/“No”) answers that indicate the presence or absence of a symbol.

Table 4.5 presents representative example prompts used for both OpenCLIP and Moondream models in this study. These examples illustrate how prompts differ in phrasing to leverage the specific strengths of each model.

Model	Example Prompt
OpenCLIP	“A black swastika symbol with arms bent at 90 degrees, typically rotated at a 45-degree angle, often shown on a red circular background or a white circle, used during World War II by Nazi Germany.”
Moondream	“Analyse whether the image contains a black swastika symbol with arms bent at 90 degrees, typically rotated at a 45-degree angle, often shown on a red circular background or a white circle, used during World War II by Nazi Germany.”

Table 4.5: Example prompts used for zero-shot classification with OpenCLIP and Moondream.

4.5 Evaluation Metrics

To quantitatively evaluate classification performance, several standard metrics commonly employed in image classification tasks were utilized. These metrics offer a multifaceted understanding of model predictive capabilities for both binary and multi-class problems.

Let the following definitions hold:

- *TP*: True Positives
- *TN*: True Negatives

- *FP*: False Positives
- *FN*: False Negatives

The metrics are formally defined as:

- **Accuracy:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy measures the proportion of correctly classified instances among all predictions and is most informative when class distributions are balanced.

- **Precision:**

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision quantifies the model's ability to avoid false positive errors, which is critical when the cost of false alarms is high.

- **Recall:**

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall assesses the model's effectiveness in identifying all relevant positive cases, particularly important in scenarios where misclassification of positive instances carry severe consequences.

- **F1-Score:**

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-Score harmonizes precision and recall into a single metric, providing a balanced measure suitable for imbalanced class distributions.

- **Area Under the ROC Curve (AUC-ROC) and Average Precision (AP) (Optional for Binary Classification):** These are threshold-independent metrics computed only for models supporting probabilistic predictions.

- **AUC-ROC** evaluates a model's discriminative ability by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) across various thresholds:

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}$$

- **AP** summarizes the precision-recall curve into a single scalar score by averaging precision across all recall levels.

These metrics were only reported for models that support probabilistic predictions. Classifiers such as `SVC`, `SGDClassifier`, and the `VotingClassifier`, as well as models like `Moondream`, were excluded from AUC-ROC and AP evaluations due to their lack of calibrated probability outputs. Similarly, the `OpenCLIP` model was excluded because it operates using cosine similarity in a high-dimensional embedding space and does not generate confidence scores or probabilistic predictions. As such, threshold-based metrics like AUC-ROC and AP are not applicable to its zero-shot, similarity-driven inference mechanism, and including them would not yield meaningful or interpretable results.

In the **multi-class setting**, AUC-ROC and AP were not employed due to their increased complexity and reduced interpretability when generalized using one-vs-rest or one-vs-one strategies. While macro- and micro-averaged variants exist, they often obscure class-specific behavior and are less suitable for this study’s imbalanced, symbol-specific dataset. Therefore, macro-averaged precision, recall, and F1-score were adopted instead, ensuring equal contribution from all classes.

All evaluation metrics were reported per model under both binary and multi-class settings. For binary classification, accuracy, precision, recall, F1-score, and—where applicable—AUC-ROC and AP were reported. For multi-class classification, macro-averaged metrics were prioritized. Model selection was guided by validation loss, and final performance metrics were computed on held-out test sets.

5 Results

5.1 Binary Classification Results

5.1.1 Tabular Results

The results for the binary classification task are summarized in Table 5.1, which reports accuracy, precision, recall, and F1-score for each evaluated model, AUC and AP for applicable models.

Model	Accuracy	Precision	Recall	F1-Score	AUC	AP
YOLO	0.995	0.793	0.932	0.857	0.999	0.913
ResNet	0.996	0.843	0.893	0.867	0.999	0.943
EfficientNet	0.993	0.772	0.761	0.767	0.993	0.780
ViT	0.997	0.910	0.893	0.901	0.999	0.962
Moondream	0.924	0.153	0.995	0.265	-	-
OpenCLIP	0.314	0.013	0.639	0.025	-	-
OpenCLIP + Logistic Regression	0.996	0.923	0.882	0.902	1.000	0.964
OpenCLIP + SGDClassifier	0.996	0.924	0.880	0.901	-	-
OpenCLIP + KNeighborsClassifier	0.996	0.895	0.936	0.915	0.999	0.934
OpenCLIP + SVC	0.997	0.948	0.929	0.939	-	-
OpenCLIP + DecisionTreeClassifier	0.987	0.674	0.742	0.706	0.993	0.625
OpenCLIP + BaggingClassifier	0.989	0.744	0.717	0.730	0.955	0.742
OpenCLIP + AdaBoostClassifier	0.994	0.901	0.791	0.842	0.998	0.925
OpenCLIP + GradientBoostingClassifier	0.993	0.879	0.801	0.838	0.998	0.919
OpenCLIP + RandomForestClassifier	0.981	0.983	0.097	0.177	0.999	0.895
OpenCLIP + VotingClassifier	0.996	0.945	0.873	0.908	-	-

Table 5.1: Binary classification performance across different models. The best scores for each metric are highlighted in bold.

5.1.2 Visual Comparison

To facilitate a clear visual comparison of model performance on the binary classification task, bar plots were generated for all selected models, including YOLO, ResNet, EfficientNet, ViT, Moondream, OpenCLIP, and OpenCLIP combined with SVC.

Figure 5.2, Figure 5.3, Figure 5.4, and Figure 5.1 depict the performance metrics, with the y-axis of each figure representing a metric of the classification results. These visualizations highlight trade-offs among models and assist in identifying the most effective architectures.

For binary classification, ViT, YOLO, and OpenCLIP-based models paired with classical classifiers—particularly SVC and KNeighborsClassifier—demonstrated superior performance. Specifically, ViT and OpenCLIP + SVC achieved the highest F1-scores of 0.901 and 0.939, respectively. Conversely, Moondream and the base OpenCLIP model exhibited comparatively lower effectiveness.

Notably, **Moondream** significantly underperformed across all metrics. This likely stems from its architectural and training objectives: Moondream is a vision-language model optimized for tasks such as caption generation and dialogue, rather than image classification. Its decoding component, aligned with safety mechanisms, may intentionally suppress or censor content related to hate symbols—including omitting, paraphrasing, or generalizing textual outputs when prompted about Nazi iconography. Such behavior undermines reliability in downstream classification tasks where semantic precision is essential. Furthermore, its visual encoder may lack the granularity necessary to distinguish between visually similar yet semantically distinct symbols—such as swastikas versus unrelated geometric designs.

Similarly, the **OpenCLIP model alone** showed limited classification capability. As OpenCLIP is trained primarily for contrastive image-text retrieval, its raw embeddings—when used without fine-tuning or a dedicated classifier—may not form well-separated decision boundaries between symbol classes. However, when paired with classical classifiers such as **SVC** or **KNN**, performance improved substantially. This suggests that while OpenCLIP embeddings capture semantically meaningful image features, they require further downstream supervision to align with fine-grained classification objectives.

Potential reasons behind the varying performance of different models, including these observations, are further examined in the Chapter 6 (Discussion and Analysis).

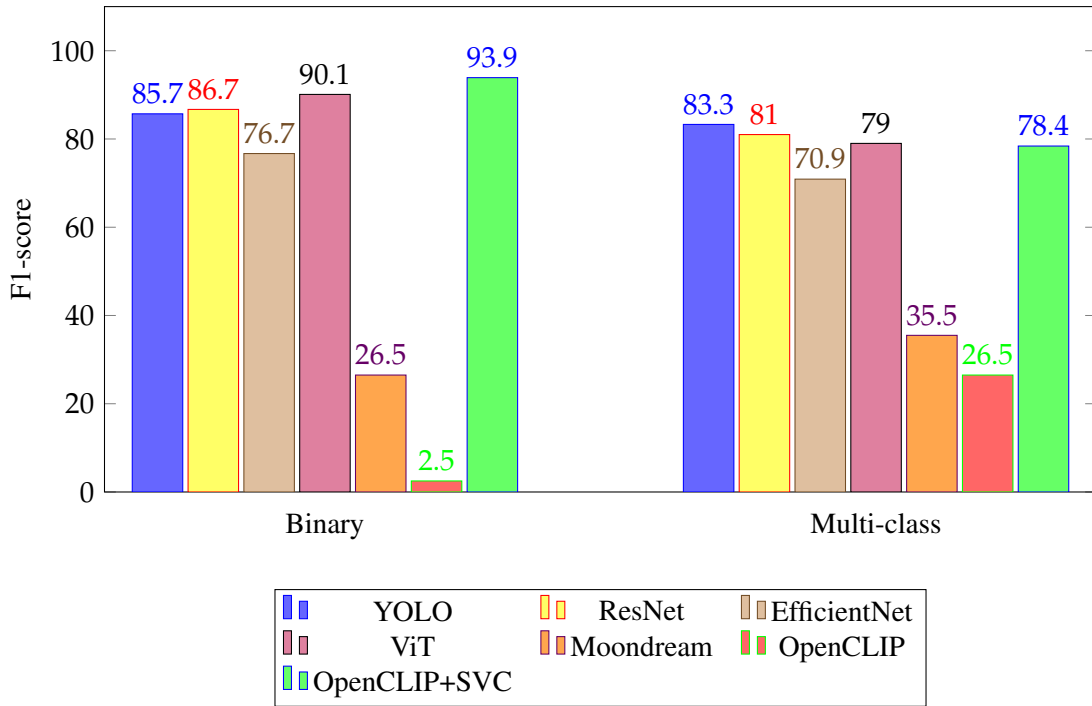


Figure 5.1: Comparison of F1-score values for all models on binary (left) and multi-class (right) classification tasks.

5.1.3 Confusion Matrices

Confusion matrices for the binary classification task are shown in Figure 5.5. Each matrix represents a model trained to distinguish images as either **Nazi** or **Non-Nazi**.

The evaluated models include YOLO, EfficientNet, ResNet, ViT, and an OpenCLIP encoder paired with a support vector classifier (SVC). These matrices visualize each model's prediction distribution by presenting true positives, true negatives, false positives, and false negatives.

In general, most models achieved strong diagonal dominance, indicating successful classification of both Nazi and non-Nazi symbols. Notably:

- **YOLO** and **EfficientNet** achieved relatively balanced true positive and true negative counts, reflecting consistent performance across both classes.
- **ResNet** and **ViT** also performed reliably, with lower false positive rates, suggesting a cautious approach that reduces mislabeling benign symbols as Nazi-related.
- The **OpenCLIP+SVC** configuration showed improved class separation compared to raw

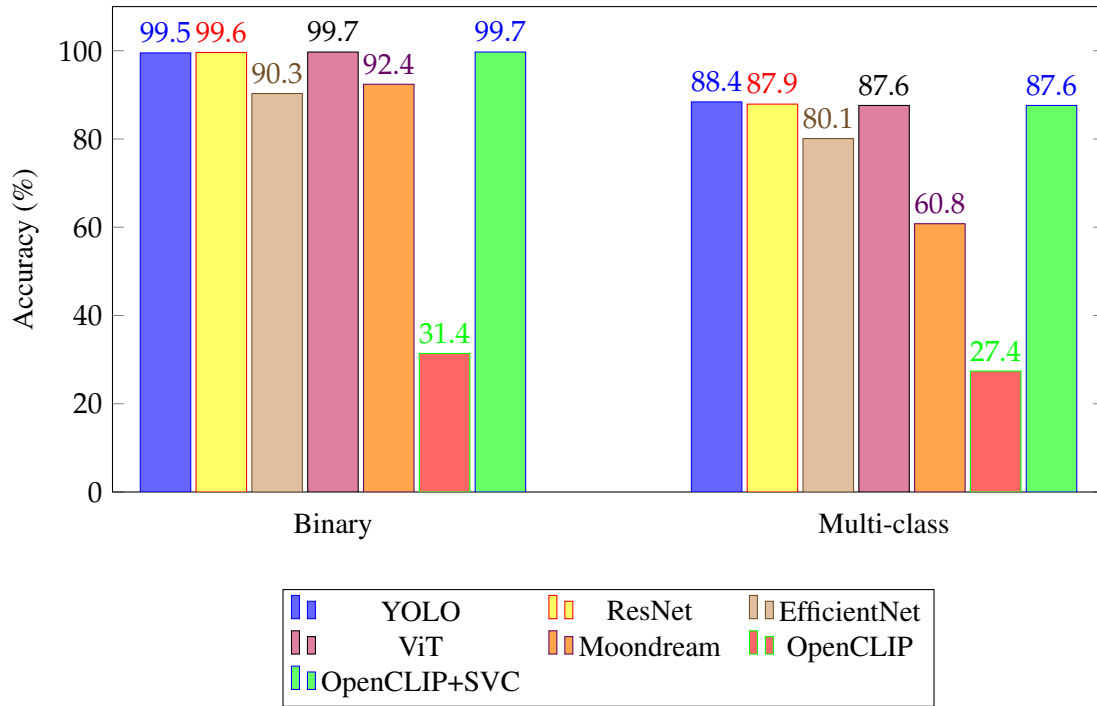


Figure 5.2: Comparison of accuracy scores for all models on binary (left) and multi-class (right) classification tasks.

OpenCLIP outputs, confirming the importance of combining feature extractors with downstream classifiers.

However, misclassifications still occurred—particularly false negatives, where Nazi symbols were not detected. These errors highlight the difficulty of classifying visually degraded, ambiguous, or stylistically subtle symbols, which will be analyzed in depth in Chapter 6 (Discussion and Analysis).

5.1.4 Receiver Operating Characteristic (ROC) and Precision-Recall (PR) Curves

To further characterize model performance under class imbalance, ROC and PR curves were plotted for the binary classification task.

Figure 5.6 illustrates the ROC curves, demonstrating the trade-off between true positive rate and false positive rate across different thresholds. The AUC values reported in Table 5.1 correspond with the visual trends observed. Most models exhibit near-perfect AUC scores (≥ 0.998), indicating strong overall separability between Nazi and Non-Nazi classes.

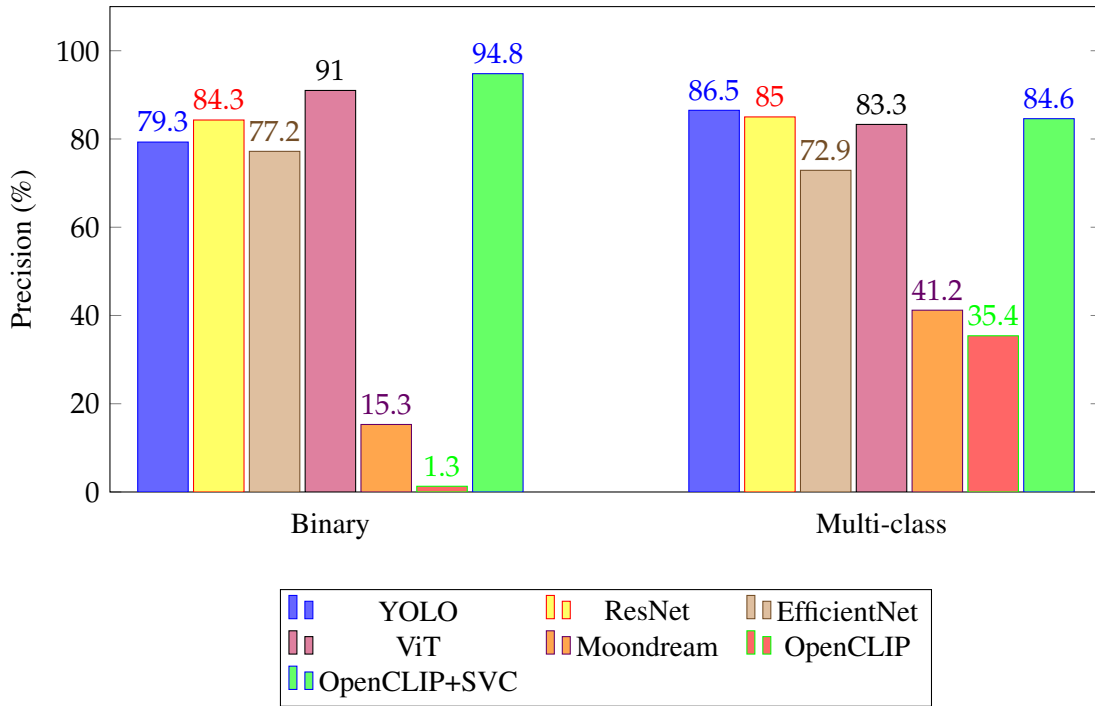


Figure 5.3: Comparison of precision scores for all models on binary (left) and multi-class (right) classification tasks.

Figure 5.7 presents the PR curves, which are particularly informative when evaluating classifiers under imbalanced conditions. The area under the PR curve (AP) complements the AUC-ROC, emphasizing each model’s ability to retrieve relevant positive samples effectively. Models such as OpenCLIP + Logistic Regression (AP = 0.964), ViT (AP = 0.962), and OpenCLIP + SVC (unreported AP but high F1) demonstrate particularly strong precision-recall trade-offs. In contrast, models like EfficientNet and DecisionTreeClassifier show notably lower AP scores (0.780 and 0.625 respectively), reflecting limitations in correctly retrieving positive instances under imbalance.

These curves and scores provide a nuanced view of classification confidence and retrieval quality beyond accuracy, particularly important in safety-critical applications where false positives or false negatives may carry different risks.

5.1.5 Reduced-Symbol Set Experiment

To further evaluate model robustness, an additional experiment was conducted using a reduced set of Nazi symbols from the original dataset. This subset included the three most visually distinct

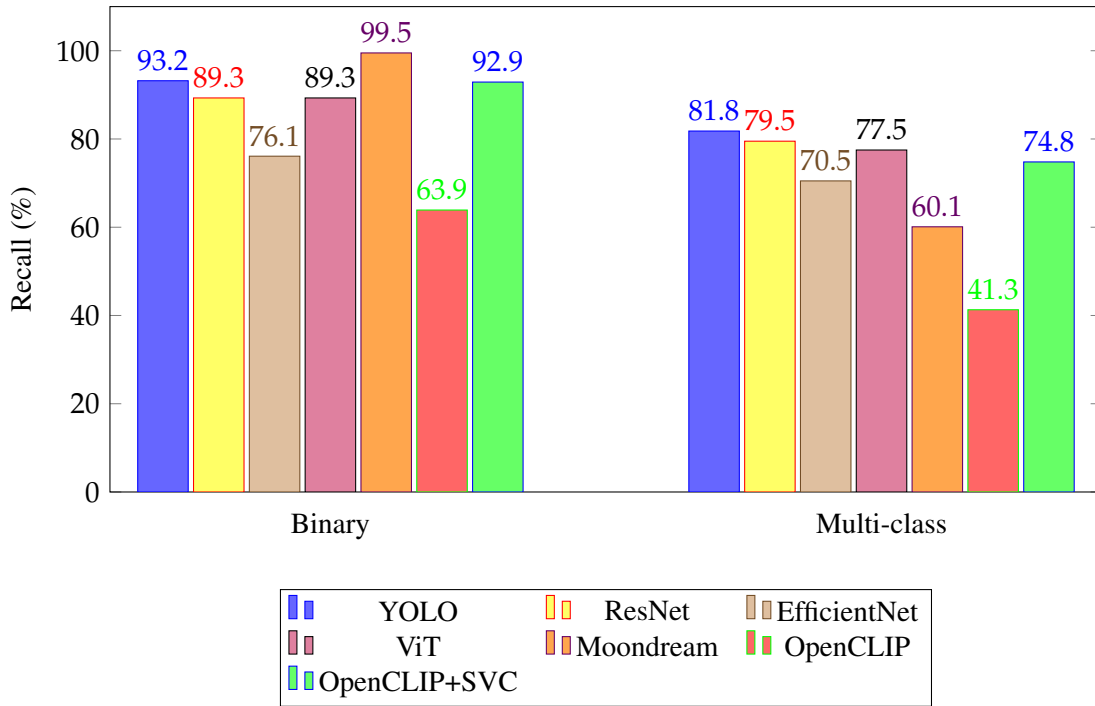


Figure 5.4: Comparison of recall scores for all models on binary (left) and multi-class (right) classification tasks.

symbols: the *SS Skull*, *Siegrune*, and *Swastika*.

The results of this reduced-symbol experiment are presented in Table 5.2, indicating the models' performance when focusing on a smaller, more visually distinct set of classes.

As expected, reducing the class set to visually distinct symbols improved performance across most models. This demonstrates the impact of inter-class visual similarity on model accuracy. With fewer symbols and more training samples per class, models could learn cleaner decision boundaries and avoid confusion between ambiguous or similar-looking symbols.

Implications of improved results on the reduced symbol set, and their relevance to dataset complexity and model robustness, are explored in more depth in the Chapter 6 (Discussion and Analysis).

Model	Accuracy	Precision	Recall	F1-Score
YOLO	0.994	0.792	0.568	0.661
ResNet	0.996	0.798	0.932	0.860
EfficientNet	0.987	0.667	0.744	0.703
ViT	0.998	0.972	0.926	0.948
Moondream	0.996	0.840	0.851	0.846
OpenCLIP	0.864	0.025	0.338	0.047
OpenCLIP + Logistic Regression	0.996	0.917	0.835	0.874
OpenCLIP + SGDClassifier	0.996	0.899	0.895	0.897
OpenCLIP + KNeighborsClassifier	0.998	0.895	0.989	0.940
OpenCLIP + SVC	0.998	0.973	0.900	0.935
OpenCLIP + DecisionTreeClassifier	0.985	0.514	0.635	0.568
OpenCLIP + BaggingClassifier	0.987	0.566	0.639	0.600
OpenCLIP + AdaBoostClassifier	0.994	0.869	0.742	0.800
OpenCLIP + GradientBoostingClassifier	0.993	0.837	0.708	0.767
OpenCLIP + RandomForestClassifier	0.988	0.959	0.209	0.344
OpenCLIP + VotingClassifier	0.997	0.964	0.833	0.894

Table 5.2: Binary classification performance across different models on reduced-symbol set. The best scores for each metric are highlighted in bold.

5.2 Multi-Class Classification Results

5.2.1 Tabular Results

The multi-class classification results are summarized in Table 5.3, reporting macro-averaged accuracy, precision, recall, and F1-score for each model.

5.2.2 Visual Comparison

Figure 5.1 Figure 5.2, Figure 5.3, and Figure 5.4 also include the multi-class classification results, presented on the right side of each plot. These visualizations allow for direct comparison of model effectiveness across multiple symbol classes.

In this setting, YOLO consistently outperformed other models across all evaluation metrics, followed closely by ResNet and ViT, both of which demonstrated strong and consistent classification performance. Models such as Moondream and OpenCLIP showed less consistent results across different classes.

5 Results

Model	Accuracy (Macro)	Precision (Macro)	Recall (Macro)	F1-score (Macro)
YOLO	0.884	0.865	0.818	0.833
ResNet	0.879	0.850	0.795	0.810
EfficientNet	0.801	0.729	0.705	0.709
ViT	0.876	0.833	0.775	0.790
Moondream	0.608	0.412	0.601	0.355
OpenCLIP	0.274	0.354	0.413	0.265
OpenCLIP + Logistic Regression	0.816	0.643	0.528	0.563
OpenCLIP + SGDClassifier	0.839	0.817	0.682	0.718
OpenCLIP + KNeighborsClassifier	0.863	0.818	0.750	0.775
OpenCLIP + SVC	0.876	0.846	0.748	0.784
OpenCLIP + DecisionTreeClassifier	0.573	0.222	0.168	0.174
OpenCLIP + BaggingClassifier	0.599	0.201	0.194	0.191
OpenCLIP + AdaBoostClassifier	0.668	0.503	0.329	0.363
OpenCLIP + GradientBoostingClassifier	0.841	0.738	0.677	0.703
OpenCLIP + RandomForestClassifier	0.573	0.246	0.169	0.170
OpenCLIP + VotingClassifier	0.836	0.757	0.557	0.609

Table 5.3: Multi-class classification performance across different models. The best scores for each metric are highlighted in bold.

5.2.3 Confusion Matrices

Confusion matrices for the multi-class classification task are shown in the figures below. Each matrix corresponds to a model trained to classify images into one of thirteen symbol categories:

```
['Black Sun', 'Flash and circle', 'Broken Sun Cross', 'The
Happy Merchant', 'Hitler', 'Nazi salute', 'Yellow badge',
'neo-Nazi', 'Siegrune', 'SS Skull', 'Sturmabteilung emblem',
'Swastika', 'Wolfsangel']
```

These matrices reveal class-specific performance, highlighting where misclassifications occur and identifying classes that are frequently confused. Such insights assist in understanding model limitations and potential areas for improvement.

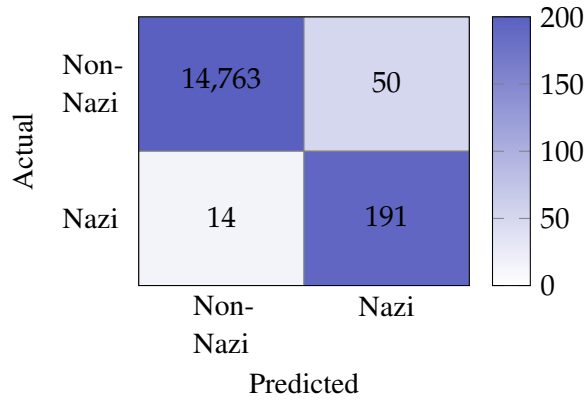
- **ResNet:** Demonstrates strong performance on prominent symbols such as *Swastika* and *Siegrune*, with occasional confusion among visually similar classes.
- **YOLO:** Achieves high accuracy on frequent classes but struggles with rarer symbols like *Nazi salute* and *Broken Sun Cross*. Overall, it has the best performance across all classes.
- **EfficientNet:** Yields balanced performance but exhibits reduced confidence on ambiguous or underrepresented categories.
- **ViT:** Shows strong recognition for dominant classes, with some misclassifications on less frequent symbols.

- **OpenCLIP encoder + SVC:** The text-image contrastive learning approach produces reasonable results, particularly for *Swastika*, but is sensitive to class imbalance.

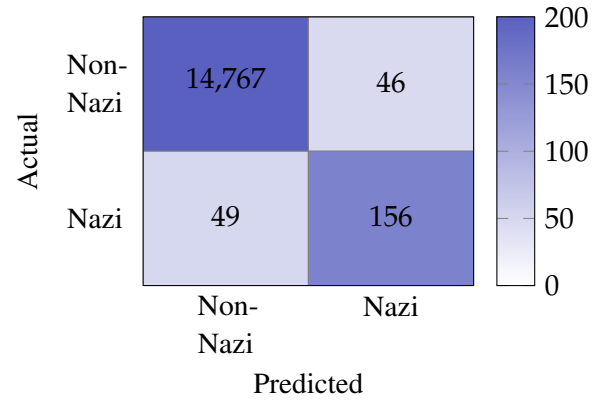
Across multi-class results, **YOLO** stands out as the top performer. Its architecture enables spatial attention and object localization, making it well-suited to distinguish between overlapping or co-located symbols. In contrast, **ViT** and **ResNet** rely on whole-image representations and struggle more with occlusion or background noise.

Moondream and vanilla OpenCLIP again performed poorly, reinforcing that general-purpose vision-language models are not optimal for fine-grained symbol classification—especially without prompt engineering or task-specific tuning.

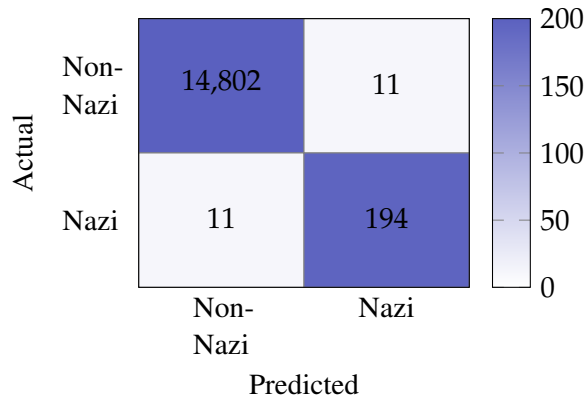
Detailed analysis and interpretation of multi-class classification challenges and model behaviors are deferred to the Chapter 6 (Discussion and Analysis).



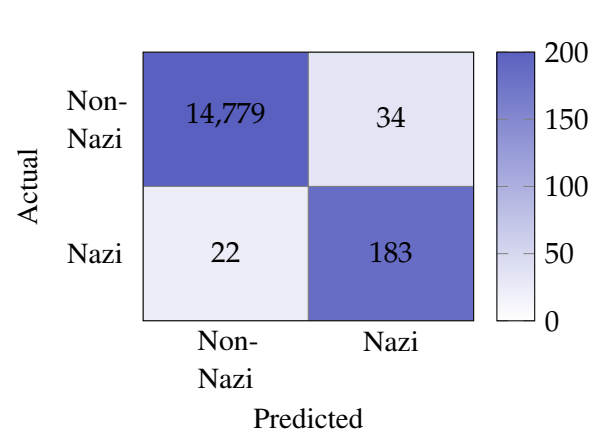
(a) Confusion matrix of YOLO.



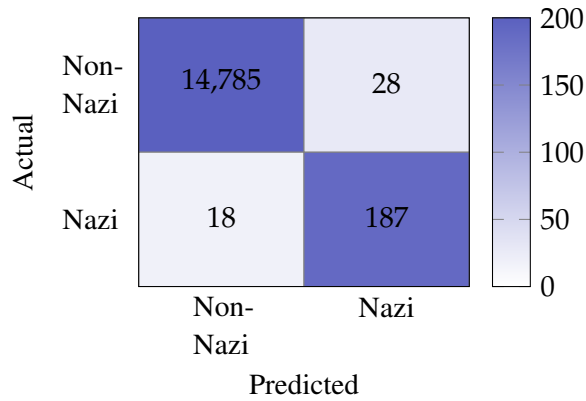
(b) Confusion matrix of EfficientNet.



(c) Confusion matrix of OpenCLIP encoder and SVC.



(d) Confusion matrix of ResNet.



(e) Confusion matrix of ViT.

Figure 5.5: Confusion matrices illustrating classification results of the binary task for each evaluated model. The matrices display true positive, true negative, false positive, and false negative counts for the classes “Nazi” and “Non-Nazi.”

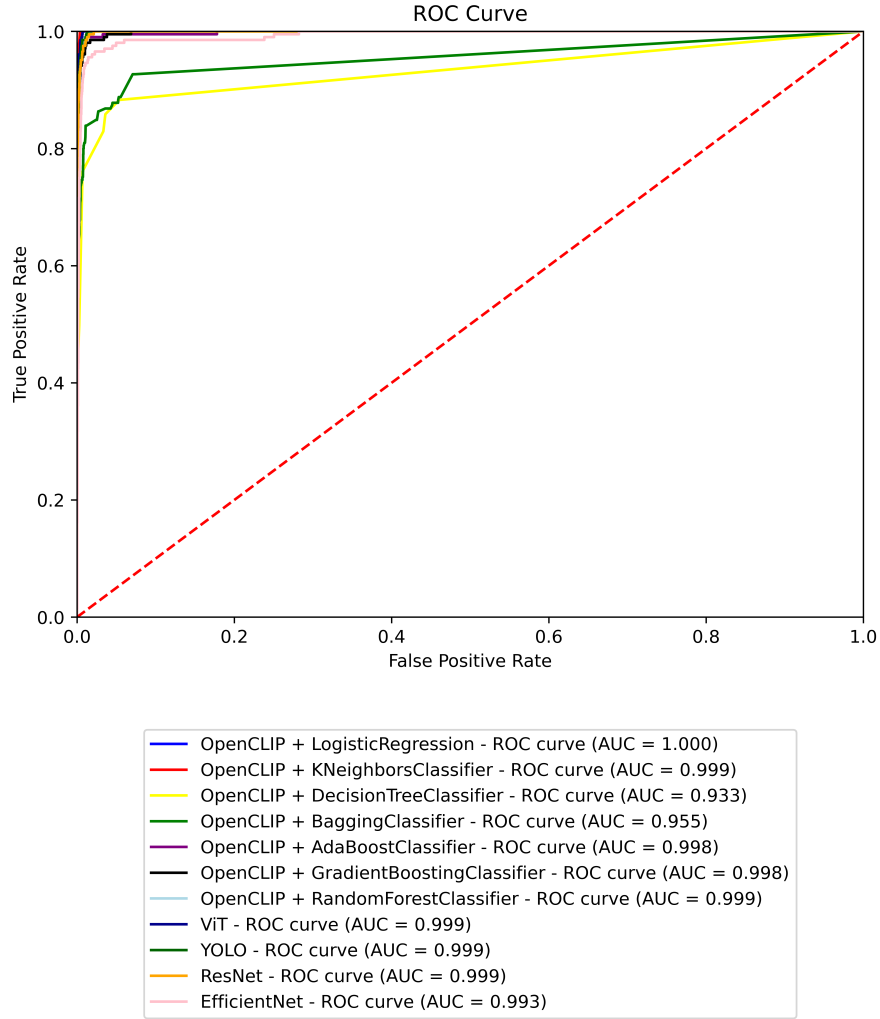


Figure 5.6: Receiver Operating Characteristic (ROC) curves for all models on the binary classification task, depicting the trade-off between true positive rate and false positive rate across classification thresholds.

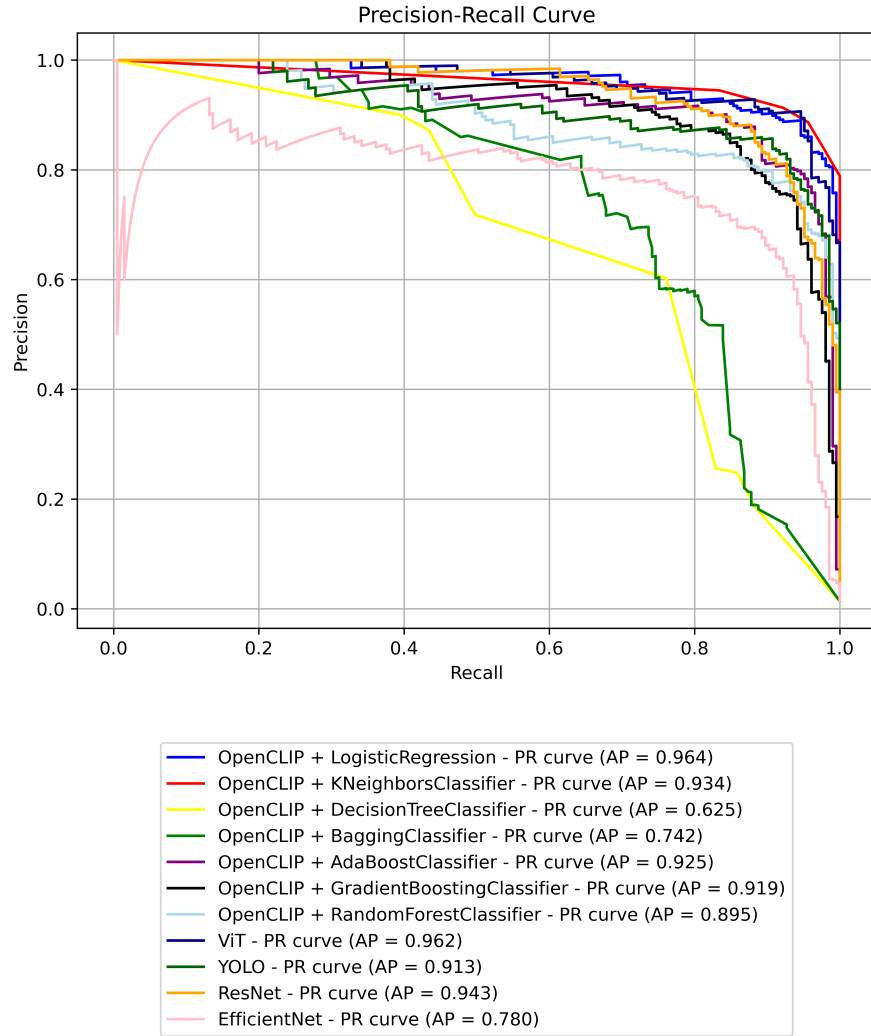


Figure 5.7: Precision-Recall (PR) curves for all models on the binary classification task, highlighting model performance under class imbalance conditions.

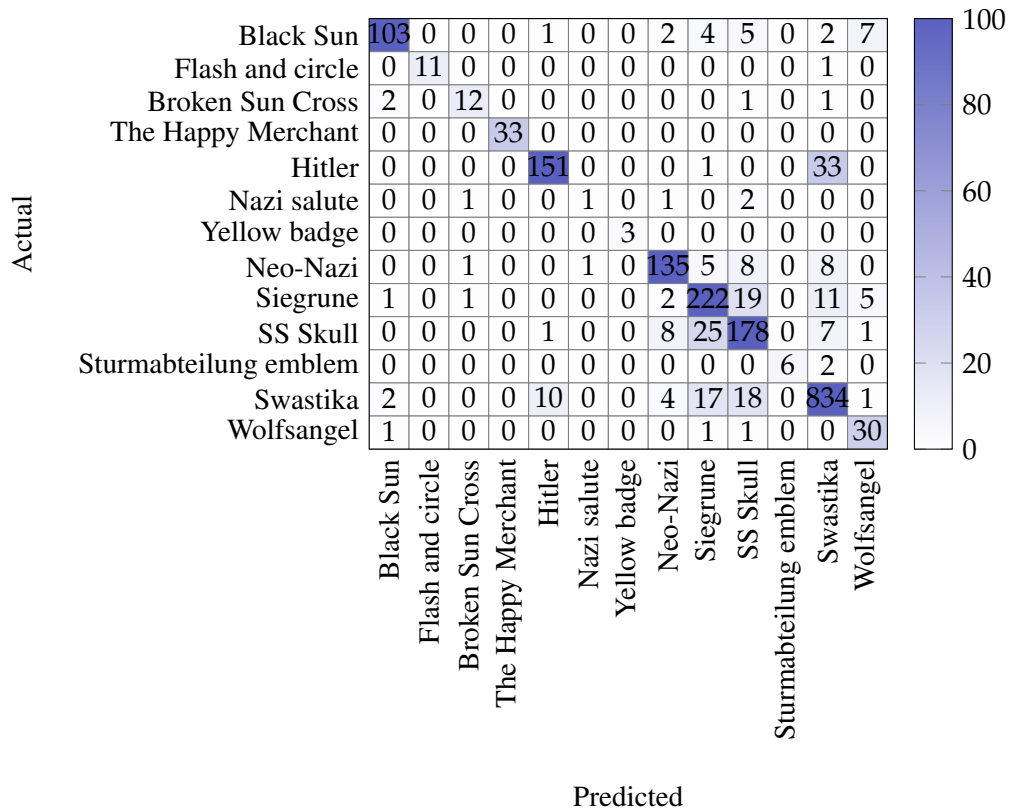


Figure 5.8: Confusion matrix for the binary classification task using YOLO. The values represent the number of samples in each cell.

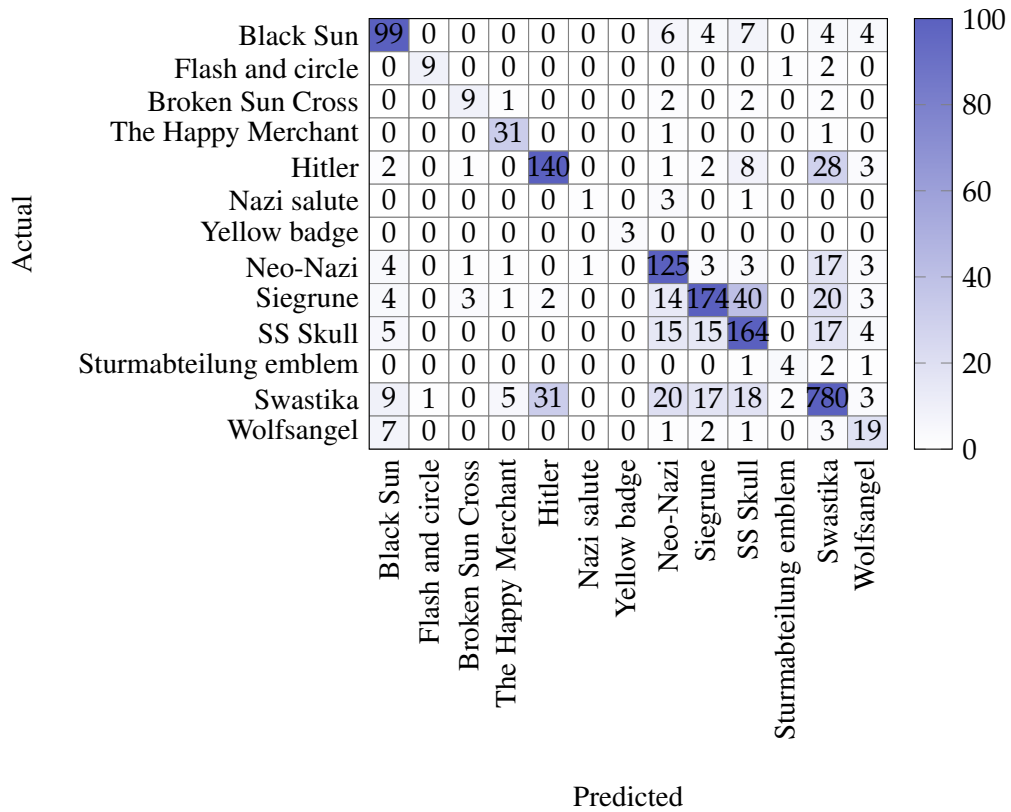


Figure 5.9: Confusion matrix for the binary classification task using EfficientNet. The values represent the number of samples in each cell.

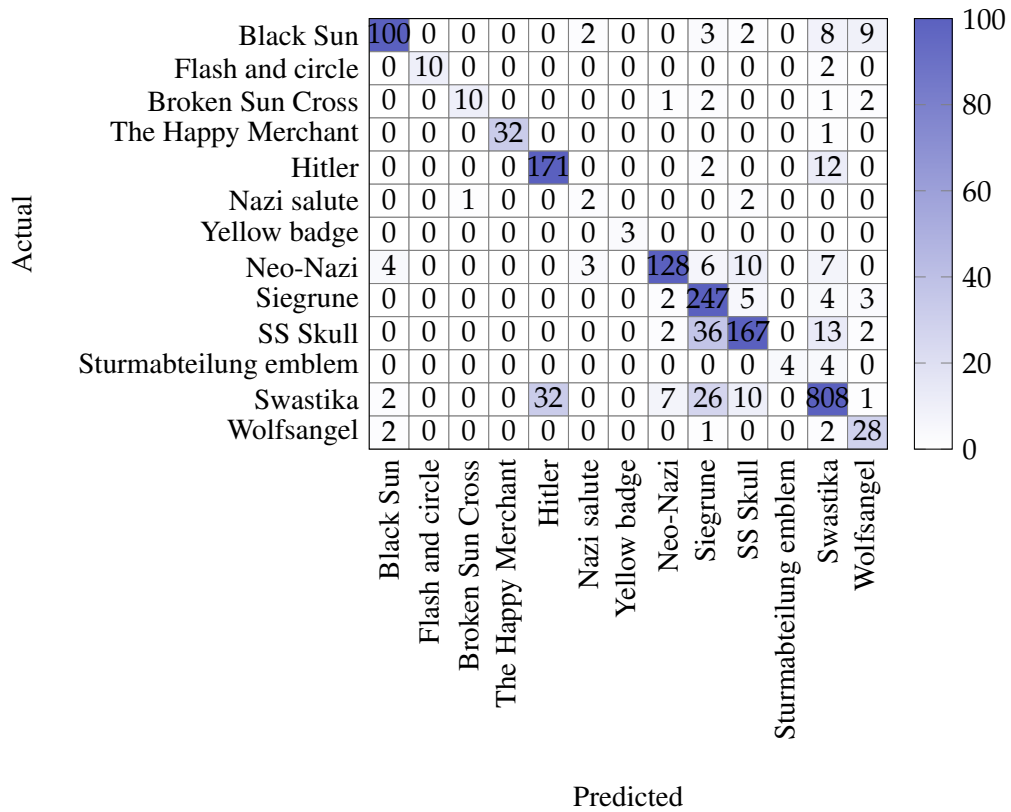


Figure 5.10: Confusion matrix for the binary classification task using ResNet. The values represent the number of samples in each cell.

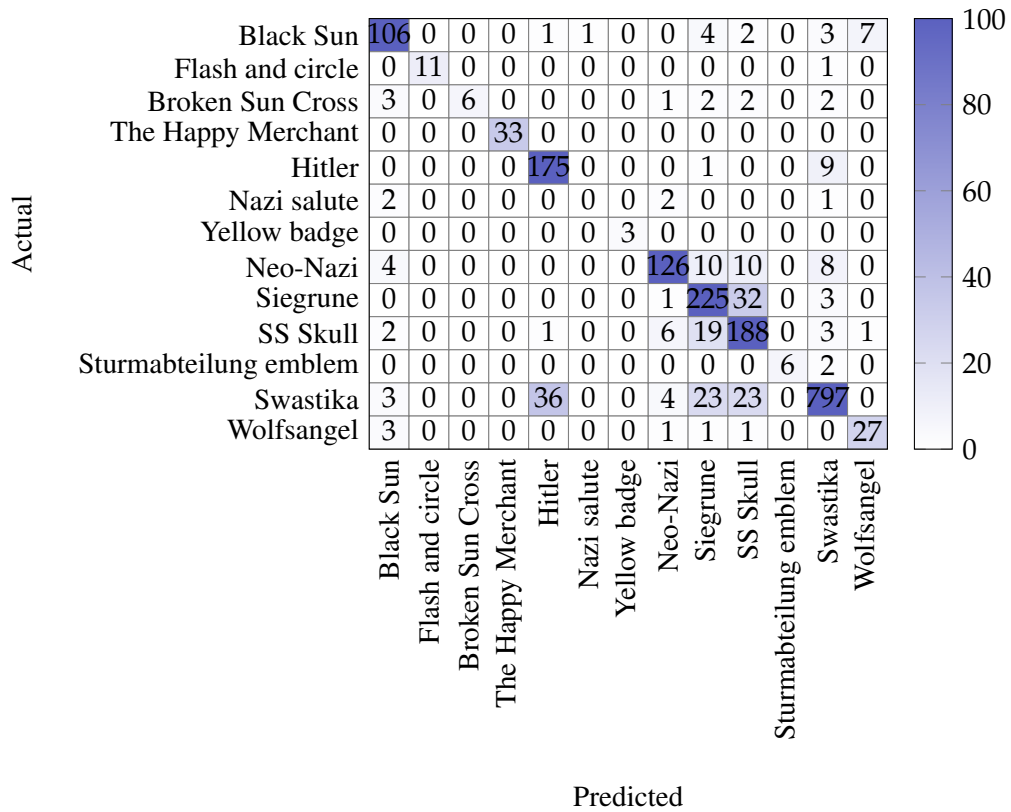


Figure 5.11: Confusion matrix for the binary classification task using ViT. The values represent the number of samples in each cell.

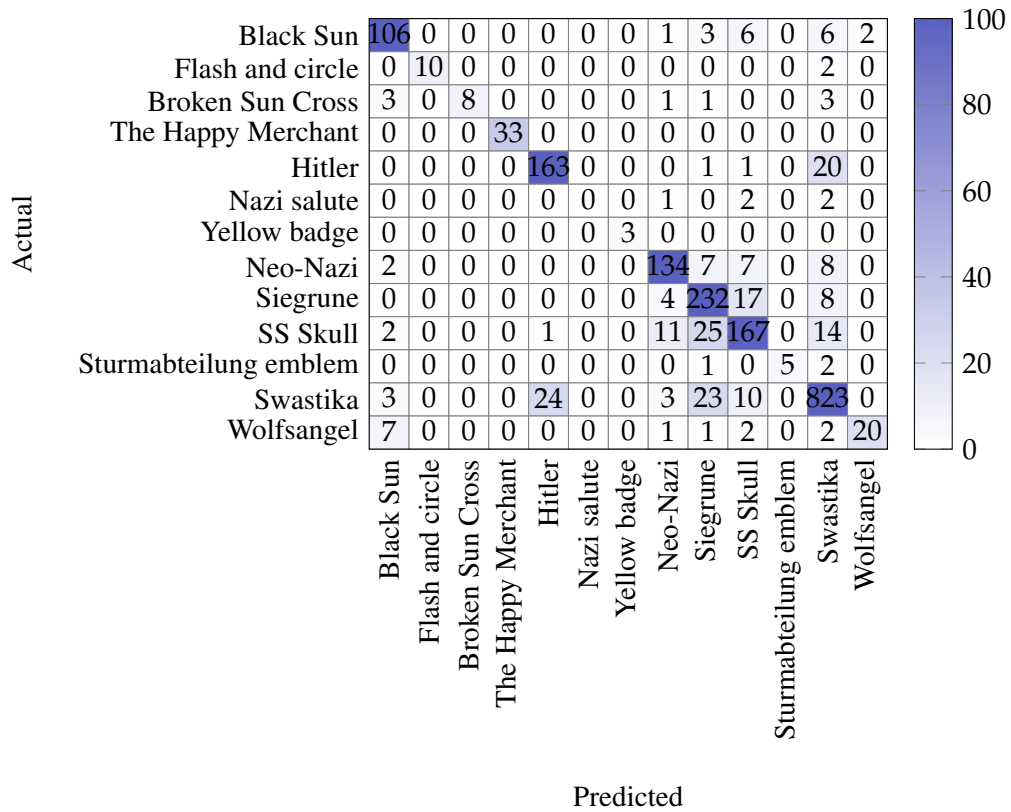


Figure 5.12: Confusion matrix for the binary classification task using OpenCLIP encoder and SVC. The values represent the number of samples in each cell.

6 Discussion and Analysis

This chapter reflects critically on the performance of the evaluated models, connecting the empirical results to the research objectives outlined in Chapter 1 (Introduction). The primary aim of this thesis was to develop and compare image classification models capable of accurately recognizing and categorizing Nazi symbols from varied visual inputs. Through detailed analysis of both binary and multi-class classification results, this chapter interprets key trends, identifies strengths and limitations of different model families, and discusses the practical and ethical implications of deploying such systems.

6.1 Analysis of Binary Classification Performance

The results presented in Table 5.1 reveal a wide range of performances across different models. Notably, classical ML classifiers leveraging OpenCLIP embeddings consistently ranked among the top performers. The best overall F1-score (0.939) was achieved by the **OpenCLIP + SVC** model, which also demonstrated high precision (0.948) and recall (0.929). Similarly, the **OpenCLIP + Logistic Regression** model achieved the highest AUC score (1.000) and highest AP (0.964), along with a strong recall of 0.882 and an F1-score of 0.902, indicating a solid ability to correctly classify Nazi symbols while minimizing false negatives.

Among DL models, both **ViT** and **YOLO** performed competitively. ViT achieved an F1-score of 0.901, the highest among all standalone neural models, while YOLO achieved an F1-score of 0.857 with a notably high recall of 0.932, highlighting its capacity to correctly classify positive symbol instances. ResNet also performed well, with an F1-score of 0.867, slightly outperforming EfficientNet (0.767).

Anomalously, the **Moondream** model achieved an exceptionally high recall of 0.995 but suffered from very low precision (0.153), resulting in a low F1-score of just 0.265. This suggests a strong tendency to overpredict the Nazi class, leading to a high number of false positives. Similarly, the **OpenCLIP zero-shot** model performed poorly, with a low F1-score of 0.025, highlighting the limitations of zero-shot classification without additional fine-tuning or classifier integration.

Several classical ensemble models also showed robust performance when paired with OpenCLIP embeddings. For instance, **OpenCLIP + VotingClassifier** reached an F1-score of 0.908 and precision of 0.945, while **OpenCLIP + AdaBoostClassifier** and **OpenCLIP + GradientBoostingClassifier** achieved F1-scores of 0.842 and 0.838, respectively. However, the **OpenCLIP**

+ **RandomForestClassifier** produced an unusually high precision (0.983) but very low recall (0.097), resulting in a poor F1-score (0.177), indicating that it was highly conservative in predicting the Nazi class.

Overall, models that combine OpenCLIP embeddings with classical classifiers—especially SVC, KNeighborsClassifier, and VotingClassifier—outperformed all other approaches in this binary classification task. Nonetheless, end-to-end deep learning models like ViT and YOLO remain competitive and demonstrate strong generalization capabilities, particularly when trained on high-resolution inputs and sufficient labeled data.

6.1.1 Binary Classification Confusion Matrix Analysis

Figure 5.5 presents the confusion matrices for the binary classification task, where the goal is to distinguish between *Nazi* and *Non-Nazi* images.

The OpenCLIP encoder combined with SVC achieves the best overall balance, producing the highest number of true positives (194) and true negatives (14,802), alongside the lowest false positives (11) and false negatives (11). This indicates a strong capability to correctly classify Nazi symbols while minimizing misclassifications.

YOLO shows excellent sensitivity, with the second lowest false negative count (14), meaning it rarely misses Nazi instances, but it has a relatively higher false positive rate (50), which might lead to more false alarms.

ResNet also performs well, showing a favorable trade-off between false positives (34) and false negatives (22), maintaining a good level of accuracy and reliability.

ViT provides balanced performance, with moderate false positives (28) and false negatives (18), showing robustness in both detecting Nazi symbols and correctly rejecting non-Nazi images.

EfficientNet, while demonstrating strong true negative results, has the highest false negatives (49), indicating that it misses more Nazi cases compared to the other models, which might limit its applicability when recall is critical.

Overall, these confusion matrices confirm that the OpenCLIP + SVC approach is the most effective for binary classification in this context, offering a strong balance between precision and recall essential for sensitive symbol classification.

6.1.2 Binary Classification ROC and PR Curve Analysis

The ROC curve results indicate that most models achieved excellent discriminative ability in the binary classification task, with several models reaching near-perfect AUC values close to 1.0. Notably, OpenCLIP combined with LR, KNeighborsClassifier, and Random Forest classifiers, along with ViT, YOLO, and ResNet, all demonstrated near-identical, near-optimal ROC performance ($AUC \approx 1.000$), suggesting their strong capacity to separate Nazi and Non-Nazi classes.

Ensemble methods such as AdaBoost and Gradient Boosting classifiers paired with OpenCLIP embeddings also performed robustly ($AUC \approx 0.998$), while EfficientNet’s slightly lower AUC of 0.993 still represents a high-quality classification boundary. Bagging and Decision Tree classifiers showed comparatively lower AUCs (0.955 and 0.933), indicating reduced robustness and potential susceptibility to misclassification in borderline cases.

The AP values from the PR curves further underscore these findings, revealing that OpenCLIP + LR and ViT provide the best balance between precision and recall, critical in scenarios where false positives and false negatives carry significant consequences. The lower AP of the OpenCLIP + Decision Tree classifier (0.625) suggests that, despite reasonable AUC scores, this model struggles to maintain precision at higher recall levels, reflecting less reliable positive class predictions.

These combined results imply that embedding-based methods (OpenCLIP) paired with strong classifiers (LR, KNeighborsClassifier) or transformer architectures (ViT) are most effective for the task. In contrast, simpler or single-model approaches such as Decision Trees show limitations in practical use despite acceptable ROC metrics.

Only models that support probabilistic outputs are included in this analysis. Classifiers such as SVC, SGDClassifier, and the VotingClassifier, as well as models like Moondream and OpenCLIP, are omitted from ROC and PR curve evaluations, as discussed in Section 4.5.

6.1.3 Binary Classification with Reduced Nazi Symbol Classes

To further examine model robustness and adaptability, a binary classification experiment was conducted. This experiment was restricted to a reduced subset of Nazi symbols (*siegrune*, *swastika*, and *SS skull*) against the full set of non-Nazi images. The goal was to evaluate model performance when the semantic diversity of the positive class is reduced.

Despite the narrower scope, overall trends remained consistent with the full binary task (as showed in Table 5.2). Transformer-based architectures continued to perform strongly. ViT achieved excellent results with an F1-score of **0.948**, indicating its strong ability to generalize even with reduced symbol variety. Among OpenCLIP-based hybrid models, *OpenCLIP + SVC* achieved the best overall performance with an accuracy of **0.998**, precision of **0.973**, recall of **0.900**, and F1-score of **0.935**, while *OpenCLIP + KNeighborsClassifier* attained the highest recall at **0.989**.

Interestingly, YOLO underperformed in this setup (F1-score **0.661**) compared to its results on the full binary task, potentially due to its reliance on spatial object features that may be less effective when class variety is reduced. Conversely, models like **Moondream** and **OpenCLIP** hybrids maintained high performance, indicating that text-image aligned embeddings may still effectively capture distinguishing features in constrained settings.

As this is a focused experiment with a reduced symbol set, the evaluation was limited to the most informative and widely supported metrics—accuracy, precision, recall, and F1-score. Metrics such as AUC and AP are omitted for simplicity and consistency, especially given that not

all models provide calibrated probability outputs.

This experiment highlights that reducing the complexity of the positive class does not necessarily simplify the task uniformly for all models. While it benefits embedding-based methods and transformers, object detection models like YOLO may be affected due to loss of visual diversity.

6.2 Analysis of Multi-Class Classification Performance

Table 5.3 summarizes the performance of all evaluated models on the multi-class classification task. The **YOLO** model achieved the best performance across all macro-averaged metrics, with an accuracy of 0.884, precision of 0.865, recall of 0.818, and F1-score of 0.833. This superior performance likely stems from YOLO’s capability to extract fine-grained spatial features in high-resolution images, combined with its efficient end-to-end object-centric architecture.

The **ResNet** and **ViT** models also demonstrated strong performance, with F1-scores of 0.810 and 0.790, respectively. ResNet maintained a slightly higher accuracy (0.879) than ViT (0.876), suggesting that deep convolutional features still offer competitive advantages. Meanwhile, ViT’s transformer-based design effectively captured semantic dependencies, reinforcing its value for tasks involving subtle symbol variations.

In contrast, **EfficientNet** underperformed in this task, with an F1-score of 0.709 and a macro recall of 0.705, indicating difficulty in generalizing across all classes. The **Moondream** model also showed weak results (F1-score of 0.355), despite a moderately high recall (0.601), suggesting overprediction of certain classes.

The **OpenCLIP zero-shot** model performed poorly overall, with an accuracy of 0.274 and F1-score of 0.265. These results reinforce that zero-shot approaches struggle when applied to highly specialized tasks like Nazi symbol recognition without task-specific adaptation or supervised training.

When OpenCLIP embeddings were combined with traditional classifiers, performance varied widely. **OpenCLIP + KNeighborsClassifier** (F1-score: 0.775) and **OpenCLIP + SVC** (F1-score: 0.784) were the most effective, indicating that neighborhood-based and kernel-based methods are capable of partially exploiting OpenCLIP’s semantic features. Similarly, **OpenCLIP + SGDClassifier** and **OpenCLIP + GradientBoostingClassifier** achieved solid F1-scores of 0.718 and 0.703, respectively.

Conversely, ensemble-based models such as **RandomForest**, **Bagging**, and **DecisionTreeClassifier** performed poorly, with F1-scores around or below 0.200. These results suggest that these models may not align well with the high-dimensional structure of OpenCLIP embeddings, potentially due to overfitting or limited ability to model fine-grained inter-class distinctions.

Overall, end-to-end models such as YOLO, ResNet, and ViT clearly outperform both zero-shot approaches and OpenCLIP embedding pipelines in the multi-class setting. These findings emphasize the importance of strong visual feature extraction and task-aligned architectures when

handling classification tasks involving subtle visual differences across many categories.

6.2.1 Comparison with Binary Classification Performance

Comparing the results from the binary (Table 5.1) and multi-class (Table 5.3) classification tasks reveals consistent trends in model effectiveness, but also highlights some key differences driven by task complexity.

In both settings, **YOLO** and **ViT** emerged as top-performing DL models. However, their relative strengths became more pronounced in the binary task. For instance, ViT achieved an F1-score of 0.901 and YOLO reached 0.857 in the binary case, whereas their multi-class F1-scores dropped to 0.790 and 0.833, respectively. This drop reflects the increased difficulty of distinguishing between a larger number of visually similar symbols.

The benefit of combining **OpenCLIP embeddings with classical classifiers** was more significant in the binary setting. In that task, models like **OpenCLIP + SVC** and **OpenCLIP + KNeighborsClassifier** achieved F1-scores of 0.939 and 0.915 respectively, outperforming nearly all other approaches. However, in the multi-class scenario, while still strong (F1-scores of 0.784 and 0.775 respectively), their advantage diminished, likely due to increased inter-class ambiguity and the limitations of traditional classifiers in modeling fine-grained boundaries across many classes.

Interestingly, some ensemble models (e.g., GradientBoosting, VotingClassifier) showed reasonable binary classification performance (F1-scores around 0.840–0.910), but generally degraded in the multi-class case (F1-scores below 0.710), possibly due to overfitting or insufficient class-separation power.

The **Moondream** and **OpenCLIP zero-shot** models performed poorly in both tasks, with Moondream showing a pathological tendency to predict the positive class in binary classification (perfect recall but extremely low precision), and generally low macro F1-scores in multi-class classification. These outcomes highlight the risks of deploying zero-shot or generative models in specialized symbol recognition tasks without task-specific training.

Overall, binary classification appears more forgiving, allowing classical classifiers with pre-trained embeddings to compete closely with end-to-end DL models. In contrast, the multi-class task places a heavier burden on precise visual discrimination and inter-class separability, where deep models like YOLO and ViT retain a clear edge. This underscores the importance of matching model complexity and architecture with task granularity and class diversity.

6.2.2 Multi-Class Classification Confusion Matrix Analysis

The confusion matrices for the multi-class classification task reveal distinct patterns of model performance across classes, reflecting varying degrees of class separation and misclassification.

ResNet (see Figure 5.10) shows strong true positive rates for high-frequency classes such

as *Swastika* (808 correct predictions) and *Siegrune* (247 correct), but also notable confusion between visually similar classes. For example, *SS Skull* is frequently misclassified as *Siegrune* (36 instances), while *Siegrune* mislabels *SS Skull* less often (5 instances). This asymmetric confusion highlights the difficulty ResNet faces in discriminating fine-grained symbol variations. Low-frequency classes like *Nazi Salute* perform poorly, with very few correct predictions.

YOLO (see Figure 5.8) generally maintains competitive accuracy for dominant classes, correctly identifying 834 *Swastika* instances and 222 *Siegrune* samples. Confusion between *SS Skull* and *Siegrune* remains substantial but somewhat more balanced compared to ResNet, with 25 *SS Skull* instances misclassified as *Siegrune* and 19 misclassifications in the opposite direction. This indicates improved inter-class separation but ongoing challenges for these similar classes.

EfficientNet (see Figure 5.9) exhibits a somewhat different confusion profile, with lower misclassification of *SS Skull* as *Siegrune* (15 instances), but higher misclassification of *Siegrune* as *SS Skull* (40 instances), showing an inverse asymmetry compared to ResNet. The true positive count for *Siegrune* is comparatively lower (174), suggesting less confident classification in this region of the feature space. Low-frequency classes continue to be problematic, with *Nazi Salute* achieving only a single correct prediction.

ViT (see Figure 5.11) marginally improves performance for certain classes, such as *SS Skull* and *Hitler*, with 188 and 175 correct predictions respectively. Confusion between *SS Skull* and *Siegrune* remains, with 19 *SS Skull* samples misclassified as *Siegrune* and 32 in the reverse direction, indicating persistent challenges in distinguishing subtle differences despite the transformer architecture's strengths.

OpenCLIP Encoder combined with SVC (see Figure 5.12) achieves a balanced confusion pattern. It correctly identifies 232 *Siegrune* and 167 *SS Skull* instances. The confusion between *SS Skull* and *Siegrune* is distributed moderately (25 and 17 misclassifications respectively), showing a more symmetric error pattern compared to other models. This suggests better inter-class separation in this challenging pair, although some misclassification persists. Additionally, the model excels at maintaining tight clusters around true labels for high-frequency classes such as *Swastika* (823 correct) but struggles with low-frequency classes like *Nazi Salute*, which has zero correct predictions.

Overall, all models face difficulties with fine-grained classes like *SS Skull* and *Siegrune*, but OpenCLIP+SVC demonstrates a more balanced confusion profile. The results highlight the ongoing need to improve discrimination for visually similar symbols, especially for classes with limited training data.

6.3 Analysis of Underperforming Models

While several models demonstrated strong classification performance, it is equally important to investigate those that underperformed to understand their limitations and potential failure modes.

The **Moondream** model demonstrated systematic failure in the binary classification task, marked by an extreme trade-off between recall (0.995) and precision (0.153), yielding an overall F1-score of only 0.265. This suggests it overwhelmingly classified images as Nazi symbols, resulting in a high false positive rate. Unlike other models, Moondream is primarily designed for vision-language tasks such as captioning and dialogue generation, not fine-grained visual classification. Its generative nature and alignment safeguards may contribute to overgeneralization or suppression of sensitive content, potentially causing it to indiscriminately signal the presence of harmful symbols. In multi-class classification, its performance remained weak (F1-score: 0.355), underscoring the model’s limited visual discrimination capability. These results highlight that even models with strong multimodal language capabilities may perform poorly on classification tasks that demand high visual specificity and consistent decision boundaries—particularly when not explicitly fine-tuned for such objectives.

The **OpenCLIP zero-shot** model underperformed across both binary and multi-class classification settings. In the binary task, it achieved an F1-score of just 0.025, and in the multi-class setting, it reached only 0.265 F1-score and 0.274 accuracy. These results suggest that OpenCLIP’s generic semantic alignment—though effective for open-ended retrieval or captioning—falls short for tasks requiring precise, domain-specific visual understanding. The lack of supervised fine-tuning likely hindered the model’s ability to map subtle symbol variations to specific class labels, especially in a setting involving ideologically charged and visually nuanced symbols. Moreover, its pretrained text embeddings may not reflect the necessary granularity or cultural-historical specificity required to distinguish, for example, a stylized swastika from unrelated geometric patterns. These outcomes reinforce the critical need for task- and domain-specific adaptation when applying general-purpose foundation models to specialized classification tasks.

Several **classical ensemble models** combined with OpenCLIP embeddings, such as the **RandomForestClassifier**, exhibited counterintuitive behavior. While the binary classification results showed extremely high precision (0.983), they suffered from near-total recall failure (0.097), resulting in a poor F1-score of 0.177. This pattern suggests an overly cautious decision boundary, where the model rarely labeled images as positive—possibly due to poor generalization in the high-dimensional OpenCLIP feature space. In the multi-class setting, ensemble models such as Random Forest, Bagging, and Decision Tree failed to exceed 0.200 F1-score, likely due to their relatively shallow decision structures and limited capacity to model the complex, fine-grained relationships embedded in pretrained vision-language features. These findings indicate that while classical models can perform well in low-dimensional or well-separated feature spaces, they struggle to generalize when paired with dense, high-capacity representations unless further feature selection or dimensionality reduction is applied.

Overall, these underperformances highlight the importance of model-task alignment. Generative or zero-shot models may offer generality, but without fine-tuning, they can fail catastrophically on specific classification tasks. Likewise, conventional tree-based models may struggle to effectively utilize dense, semantic embeddings like those from OpenCLIP, especially in tasks with

fine-grained categories.

These failures provide a contrast to the following high-performing approaches, which better align with the classification objectives.

6.4 Analysis of High-Performing Models

Across both binary and multi-class classification tasks, several models demonstrated exceptional performance, providing insight into the characteristics of effective architectures for Nazi symbol recognition.

The highest-performing binary classifier was the combination of **OpenCLIP embeddings with a SVC**, which achieved an F1-score of 0.939, paired with a high precision (0.948) and recall (0.929). This suggests that the kernel-based decision boundaries of SVC are well-suited to partitioning OpenCLIP’s semantically rich, high-dimensional embedding space. Its confusion matrix reveals both low false positive and false negative rates, indicating stable and balanced performance even under class imbalance. Similarly, the **OpenCLIP + LR** model achieved the best AUC (1.000) and AP (0.964), highlighting its strength in probabilistic calibration. The fact that even a linear model performs so well when paired with OpenCLIP embeddings underlines the quality of these pretrained representations, which already encode strong semantic separability between Nazi and non-Nazi symbols. Together, these results demonstrate that lightweight classifiers—when paired with powerful pretrained vision-language embeddings—can deliver highly competitive performance on specialized tasks, without requiring deep end-to-end training.

In the multi-class classification setting, **YOLO** emerged as the top-performing model, achieving a macro F1-score of 0.833. Unlike purely classification-based architectures, YOLO’s detection-oriented structure inherently emphasizes spatial localization and region-level features, which may be especially advantageous for distinguishing fine-grained visual details among similar-looking symbol classes. Its strong per-class performance on high-frequency symbols like *Swastika* and *Siegrune* suggests it effectively captures both global and local patterns. Moreover, YOLO’s anchor-based detection approach may offer a form of implicit segmentation, aiding the model in isolating relevant symbol regions from noisy or cluttered image backgrounds—a recurring challenge in real-world deployment contexts.

Both **ResNet** and **ViT** demonstrated consistently strong performance across binary and multi-class tasks, with F1-scores above 0.79 in all settings. ResNet’s deep convolutional layers facilitated robust extraction of low- to mid-level visual features, while its residual connections likely improved gradient flow and optimization stability during training. ViT, in contrast, relies on a patch-based self-attention mechanism, which enables it to model long-range dependencies and complex compositional structures—a key advantage when symbols contain subtle stylistic variations. Their strong multi-task performance reinforces the effectiveness of end-to-end deep learning architectures when trained on sufficient, well-labeled domain data. These models serve

as reliable baselines for further exploration of task-specific adaptations or multimodal extensions.

Among embedding-based approaches, **OpenCLIP + KNeighborsClassifier** also stood out with strong binary (F1-score: 0.915) and competitive multi-class (F1-score: 0.775) performance. The effectiveness of KNN suggests that OpenCLIP embeddings preserve semantic distances in a way that aligns well with class membership—enabling nearest-neighbor comparisons to retrieve visually and semantically similar symbols. This model’s strength may also stem from its non-parametric nature, which avoids assumptions about the shape of decision boundaries and instead relies on local geometric structure in the embedding space. Such behavior could be particularly valuable in classification settings where labeled examples are sparse but clustered, and where explainability through exemplar retrieval is desirable.

Across high-performing models, a common theme emerges: the combination of rich visual representations—whether through end-to-end convolutional or transformer architectures, or pretrained embeddings like OpenCLIP—with well-matched classification strategies, yields robust symbol classification. These results emphasize the value of pairing strong feature extractors with appropriately complex decision functions, tailored to the geometry of the input space and the semantic structure of the task.

Beyond individual model metrics, broader themes emerge across tasks and models, offering insight into effective strategies for symbol classification.

6.5 Key Insights and Thematic Patterns

Analysis across both binary and multi-class tasks revealed several recurring patterns that shed light on the design trade-offs and performance dynamics of different model classes.

Hybrid architectures outperform raw foundation models. One of the most consistent findings was the superior performance of OpenCLIP embeddings when combined with classical machine learning classifiers such as SVC, LR, or KNN. These hybrid models outperformed both traditional CNNs and foundation models used in isolation (e.g., zero-shot OpenCLIP or Moondream). The results suggest that while pretrained vision-language embeddings capture rich semantic structure, they require tailored downstream classifiers to translate these representations into accurate symbol predictions. This modular approach also offers flexibility in tuning precision-recall trade-offs and model interpretability.

Transformer-based models excel in nuanced visual tasks. Vision Transformer (ViT) models demonstrated consistently strong results, benefiting from global self-attention mechanisms that allow for nuanced discrimination between visually similar symbols. Compared to CNNs like ResNet or EfficientNet, ViT showed greater robustness to stylization, partial occlusion, and background noise. This suggests that transformer architectures may be better suited for iconographic

classification tasks that require compositional understanding and spatial context awareness.

Model architecture impacts suitability for deployment. The classification-specific variant of the YOLO model—developed by the YOLO team—achieved the best performance in the multi-class classification task, with the highest accuracy (0.884), precision (0.865), recall (0.818), and F1-score (0.833). These results highlight the strength of streamlined, efficient architectures in real-world settings where both performance and inference speed matter. Unlike traditional object detectors, this version of YOLO is natively optimized for classification tasks. Its ability to maintain SotA accuracy while remaining computationally lightweight makes it especially well-suited for deployment in constrained environments or applications requiring real-time symbol recognition.

Generalization to historical and stylized symbols remains a challenge. Several models struggled when presented with visually degraded, stylized, or low-contrast images typical of historical archives or user-generated content. Despite strong performance on curated datasets, generalization gaps remain, particularly for rare or obscure Nazi symbols. This points to the need for domain adaptation strategies such as contrastive pretraining, synthetic augmentation, or inclusion of diverse training sources to enhance model robustness.

Symbol frequency imbalance affects recall and stability. As in many multi-class classification settings, class imbalance significantly impacted model performance. Rare symbol classes, such as the *Black Sun*, exhibited low recall and unstable predictions, even in otherwise high-performing models. Addressing this will likely require targeted techniques such as few-shot learning, oversampling, or curriculum-based training to ensure equitable performance across all classes.

Together, these insights highlight the trade-offs between model complexity, interpretability, and task suitability—informing how future systems can balance accuracy with practicality and ethical deployment needs.

6.6 Implications for Real-World Deployment

The findings suggest that hybrid approaches—combining deep feature extraction with classical ML classifiers—offer the most promise for deployment in real-world systems tasked with binary classification of Nazi symbols in images. In particular, OpenCLIP combined with SVC consistently achieved the highest F1-score in the binary classification task, indicating strong performance in balancing precision and recall when identifying rare and sensitive symbols. These hybrid models also offer a favorable trade-off between accuracy, interpretability, and computational efficiency—key considerations for safety-critical applications such as content moderation or archival screening.

However, practical deployment requires addressing several challenges. These include handling ambiguous or degraded images, unseen symbol variants, and the high cost of false positives in legal, historical, or moderation contexts. The results suggest that a human-in-the-loop approach may be appropriate, where automated classification is supplemented by expert review for uncertain or high-stakes cases.

Moreover, compliance and interpretability are critical. Simpler classifiers layered on OpenCLIP embeddings offer greater transparency than end-to-end deep networks, which can support auditability, regulatory acceptance, and user trust.

These deployment-oriented considerations directly support the research objective of developing practical, accurate, and ethically responsible classification systems for real-world symbol recognition tasks.

6.7 Limitations and Risks

Despite promising results, several limitations and risks remain. Most notably, model performance on minority classes was inconsistent, with a tendency to overfit to frequent, visually dominant symbols. This raises concerns around fairness, completeness, and the system’s ability to generalize to underrepresented symbol types in deployment.

Dataset bias also presents a notable risk. Much of the training data was synthetically curated or drawn from known repositories, potentially omitting rare, obscure, or modern reinterpretations of Nazi iconography. As a result, the system may misclassify or entirely miss symbols that deviate from the curated training distribution.

Another key limitation is the lack of contextual awareness. The models operate solely on visual information and do not distinguish between educational, artistic, or propagandistic usage. While context modeling was beyond the scope of this project, it is essential for responsible deployment. Without it, there is a risk of misinterpretation, overreach, or harm—particularly in sensitive social, legal, or historical settings.

These limitations highlight the need for further work in data diversification, contextual modeling, and ethical evaluation before deploying such systems at scale.

7 Model Deployment and API Service

To facilitate real-world application and integration of the trained classification models, an API service was developed to serve the best-performing models from both the binary and multi-class classification tasks. The service was designed with a strong emphasis on modularity, portability, and usability.

Although the API was not deployed to an online hosting platform due to infrastructure constraints, it was fully implemented, containerized using Docker, and published to Docker Hub. This approach ensures reproducibility and allows collaborators (e.g., from the social sciences) or third-party researchers to easily run the service in local or cloud-based environments supporting Docker runtime.

7.1 System Architecture

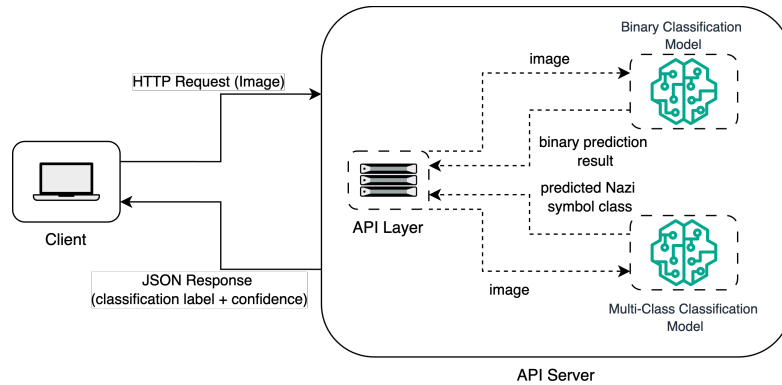


Figure 7.1: System architecture of the implemented classification API service.

Figure 7.1 illustrates the system architecture of the classification API service. Users submit an image file via an HTTP request, which is routed to one of three model pipelines depending on the specified task: binary classification, multi-class classification, or a two-stage hierarchical classification pipeline.

In the two-stage pipeline, the image is first passed to a binary classifier to determine whether it contains Nazi symbolism. If no such content is detected, the API returns a negative classification.

If Nazi-related content is identified, the image is subsequently forwarded to the multi-class classifier, which predicts the specific type(s) of Nazi symbol(s), such as swastikas, SS bolts, or eagles.

This cascaded design reduces computational overhead for irrelevant inputs and mirrors real-world content moderation workflows, where fine-grained analysis is only triggered upon initial positive screening.

The backend is implemented using `FastAPI`. All models are preloaded into memory at service startup, ensuring low-latency inference. The architecture is stateless and fully containerized, enabling straightforward horizontal scaling in distributed environments.

7.2 Model Integration and Routing Logic

The API exposes distinct endpoints for binary classification, multi-class classification, and the combined 2-layer pipeline. Instead of selecting the model dynamically based on request parameters, each endpoint corresponds to a specific inference path.

This design simplifies both the server-side routing and the client interface, enabling users to directly interact with the appropriate endpoint. Each endpoint internally handles preprocessing, model inference, and result formatting.

At service startup, all models are preloaded into memory. DL models implemented with PyTorch are serialized using `TorchScript`¹ or the standard PyTorch save format². Classical ML models (e.g., logistic regression or SVC trained on OpenCLIP embeddings) are serialized using `joblib`³. This preloading strategy reduces runtime latency and avoids repeated I/O overhead.

The modular architecture allows easy extension. New models or task-specific pipelines can be integrated by registering additional routes in the API codebase, without disrupting existing functionality.

Figure 7.2 summarizes the endpoint-to-model routing logic.

7.3 Continuous Integration and Deployment

The service is integrated with a GitHub Actions-based CI/CD pipeline, shown in Figure 7.3, which automates:

- Code quality checks and unit testing
- Docker image builds and tagging

¹<https://pytorch.org/docs/stable/jit.html>

²https://pytorch.org/tutorials/beginner/saving_loading_models.html

³<https://joblib.readthedocs.io/en/latest/>

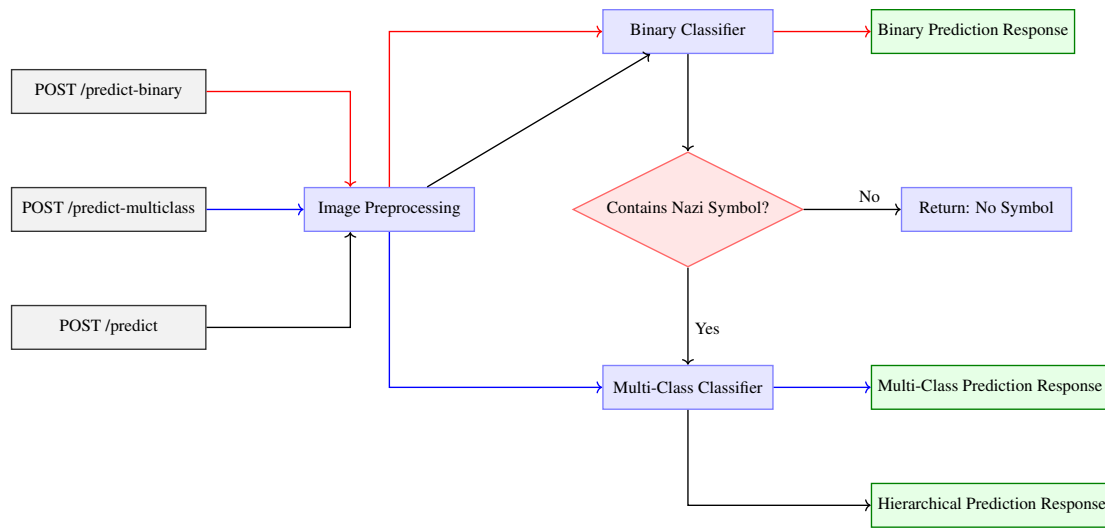


Figure 7.2: API endpoints mapped to model pipelines, including the 2-layer classification logic.

- Generation and deployment of Sphinx-based documentation
- Publication of Docker images to Docker Hub

Although the API is not deployed to a public endpoint, the pipeline ensures that each commit to the main branch undergoes validation and build procedures, promoting maintainability. Users can reliably pull the latest version from Docker Hub and execute the service locally or in private cloud environments with minimal setup.

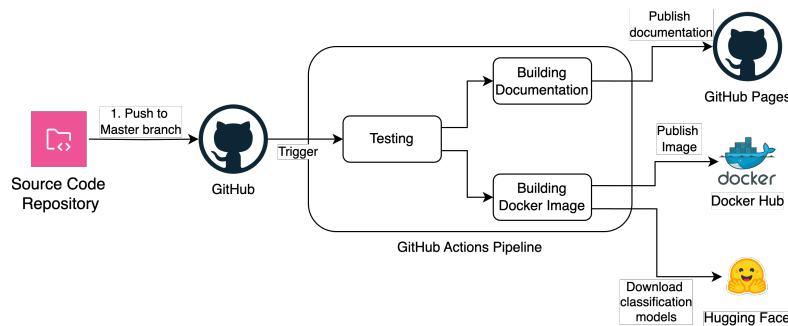


Figure 7.3: GitHub Actions CI/CD pipeline for Dockerized API service.

7.4 API Endpoints

Table 7.1 summarizes the available endpoints and their functionality.

The 2-layer endpoint encapsulates the binary–multi-class cascade, simplifying client-side logic and supporting integration into broader moderation or research pipelines.

Method	Endpoint	Description
POST	/predict	Runs the 2-layer pipeline: binary classification followed by multi-class prediction if Nazi content is detected. Threshold for symbol inclusion is 0.1.
POST	/predict-binary	Performs binary classification with associated confidence scores.
POST	/predict-multiclass	Performs multi-label classification of Nazi symbol types. Returns predictions above confidence threshold 0.1.
GET	/health	Health check endpoint.
GET	/version	Returns model and API version metadata.
GET	/docs	Serves OpenAPI documentation via Swagger UI.

Table 7.1: Summary of available API endpoints.

All classification endpoints accept images as multipart form data and return structured JavaScript Object Notation (JSON) responses. These responses include predicted classes, confidence scores, and model metadata. The interactive Swagger UI at `/docs` facilitates testing and exploration.

7.4.1 Example JSON Response

Figure 7.4 shows a sample JSON response for the binary classification endpoint. Figure 7.5 illustrates a multi-class classification response, while Figure 7.6 demonstrates the 2-layer classification response format.

```

1 [
2   {
3     "prediction": "nazi",
4     "confidence": 0.965
5   }
6 ]

```

Figure 7.4: Sample JSON response for binary classification.

```
1 {
2   "nazi_symbols": ["swastika", "siegrune", "black_sun"],
3   "details": [
4     {"label": "swastika", "confidence": 0.882},
5     {"label": "siegrune", "confidence": 0.062},
6     {"label": "black_sun", "confidence": 0.016}
7   ]
8 }
```

Figure 7.5: Sample JSON response for multi-class classification.

```
1 {
2   "containing_nazi_symbols": true,
3   "prob": 0.97,
4   "nazi_symbols": ["swastika", "siegrune", "black_sun"],
5   "details": [
6     {"label": "swastika", "confidence": 0.882},
7     {"label": "siegrune", "confidence": 0.062},
8     {"label": "black_sun", "confidence": 0.016}
9   ]
10 }
```

Figure 7.6: Sample JSON response for 2-layer classification.

7.4.2 Thresholding vs. Top- k in Multi-label Classification

In the multi-class classification task, images may contain more than one Nazi symbol, constituting a multi-label problem. Instead of using a top- k strategy, which returns the k most confident classes regardless of absolute probability, a fixed confidence threshold of 0.1 was used to filter results.

This design offers two key advantages:

1. **Adaptability to Varying Label Counts:** The number of relevant symbols varies across images. A threshold-based method allows the model to return only those classes exceeding the confidence threshold, thereby adapting to the input's true label count.
2. **Reduction of False Positives:** Low-probability predictions are filtered out, reducing the likelihood of returning irrelevant or incorrect classifications. This is particularly important in sensitive domains such as Nazi symbolism detection, where misclassification could have serious social and legal implications.

This approach provides more nuanced, context-aware predictions and aligns with the goal of responsible content moderation.

7.5 Extensibility and Future Deployment

The API service is architected for extensibility. Future capabilities—such as object detection for symbol localization, Optical character recognition (OCR) for embedded text, or detection of other hate symbols—can be integrated by adding new model pipelines and routes. The decoupled routing logic and memory preloading facilitate straightforward extensibility.

Although the service is not deployed online, it is fully containerized and ready for local or remote deployment. The accompanying GitHub repository includes:

- A `Dockerfile` and `pyproject.toml` for consistent runtime environments
- Instructions for local deployment using `docker run` or `docker-compose`
- Example requests (e.g., via `curl`) and a Postman collection
- GitHub Actions workflows for testing, documentation generation, and Docker publishing

This setup supports reproducibility and enables non-technical collaborators (e.g., from social sciences) to run the service locally without additional configuration.

For future deployment on public platforms, the system could be hosted on AWS, Google Cloud, or Azure. It is compatible with container orchestration (e.g., Kubernetes) or serverless deployment (e.g., AWS Lambda + API Gateway). Additional features—such as user authentication, request throttling, and secure logging—would be needed for production-grade deployment.

The modular and portable design ensures that the service can evolve alongside the research project and scale into production or collaborative use cases as required.

8 Conclusion and Future Work

8.1 Conclusion

This thesis presented a comprehensive study on classifying Nazi symbols in images, exploring a wide range of architectures—including CNNs (ResNet and EfficientNet), transformer-based models (ViT), classification-specific versions of YOLO, and multimodal approaches leveraging OpenCLIP embeddings. In addition to end-to-end deep learning pipelines, hybrid approaches that combined OpenCLIP features with classical ML classifiers were also evaluated.

Two classification settings were explored: binary classification (distinguishing Nazi from non-Nazi symbols) and multi-class classification across 13 Nazi-related symbol categories.

In the binary task, hybrid models—particularly OpenCLIP paired with classical classifiers such as SVC, LR, KNeighborsClassifier, and SGDClassifier—consistently delivered high performance, achieving F1-scores above 0.89 and, in some cases, perfect ROC AUC scores (e.g., LR with AUC = 1.00). Transformer-based ViT and classification-oriented YOLO also demonstrated strong results, balancing accuracy, precision, and recall. These findings affirm the value of both pre-trained visual-semantic embeddings and end-to-end deep feature extractors for sensitive symbol classification.

Nonetheless, some hybrid configurations—such as OpenCLIP combined with Decision Tree or Bagging Classifiers—underperformed, especially in PR metrics, signaling limited robustness under class imbalance. Similarly, EfficientNet achieved high AUC (0.993) but lower AP (0.780), making it less suitable for applications requiring fine-grained discrimination.

In the multi-class setting, YOLO emerged as the top performer, achieving the highest accuracy, precision, and F1-score. ViT also performed strongly, confirming the effectiveness of global attention mechanisms for nuanced symbol differentiation. While OpenCLIP performed poorly in zero-shot mode, its performance improved significantly when combined with supervised classifiers—though still behind end-to-end models. These results highlight the importance of downstream supervision when applying foundation models to fine-grained classification tasks.

In summary, this work demonstrates that both deep learning and hybrid approaches can be effective for the classification of Nazi symbols. The choice of model should be guided by task constraints, desired performance metrics, and deployment requirements. Success ultimately depends on a balance between representational power, classifier robustness, and sensitivity to class imbalance and symbol diversity—critical considerations for building practical and ethically responsible systems.

8.2 Future Work

Building on the encouraging results of this study, several key directions for future research emerge:

Dataset Expansion and Balancing: Increasing the diversity and quantity of labeled images—particularly for rare or visually ambiguous Nazi symbols—could significantly improve performance in the multi-class setting. Incorporating hard negatives (e.g., geometric symbols or cultural motifs that resemble Nazi iconography) would also enhance binary classifier robustness against false positives.

Fine-Tuning Vision Backbones and Multimodal Retraining: While OpenCLIP embeddings paired with classical classifiers showed strong performance, they were limited in zero-shot generalization. Task-specific fine-tuning of the OpenCLIP vision encoder—or end-to-end multimodal retraining with labeled Nazi symbol data—may lead to better feature separability and improved recall across all classes.

Precision-Focused Learning Objectives: As shown in the results, high AUC did not always translate into high real-world precision. Future work could experiment with cost-sensitive loss functions, confidence calibration, or threshold tuning to explicitly optimize for precision-recall trade-offs—especially important in applications with legal or reputational risks.

Explainability and Human Oversight: Integration of explainable AI (XAI) tools such as Gradient-weighted Class Activation Mapping (Grad-CAM) [Sel+19] or SHapley Additive exPlanations (SHAP) [25] would improve transparency and accountability. Providing visual or textual explanations for predictions could support expert review and increase user trust—especially in archival or forensic applications.

Contextual and Multimodal Classification: Some misclassifications were driven by lack of context. Incorporating auxiliary metadata (e.g., captions, filenames, web text) and exploring multimodal fusion models could improve classification of ambiguous or reused symbols, and help distinguish educational versus propagandistic usage.

Real-Time Deployment and System Scalability: Future engineering work should extend the current containerized API toward production-ready deployment. Key goals include low-latency inference, cross-platform scalability, and compliance with privacy regulations (e.g., GDPR), making the system usable in content moderation or institutional archiving workflows.

Human-in-the-Loop Feedback and Adaptation: Building mechanisms for expert feedback and iterative improvement would support continuous learning. This is especially valuable in evolving symbol ecosystems where new variants or context-specific uses emerge over time.

In summary, future efforts should aim to enhance the system’s precision, transparency, and adaptability to sensitive real-world environments. By combining improved data practices, targeted fine-tuning, explainability, and user input, the classifier can evolve into a more trustworthy and context-aware tool for ethical Nazi symbol recognition and moderation.

Abbreviations

ViT Vision Transformer

LR Logistic Regression

CNN Convolutional Neural Network

ML Machine Learning

CI/CD Continuous Integration/Continuous Delivery

DL Deep Learning

ZSL Zero-Shot Learning

CLIP Contrastive Language-Image Pre-training

RN Relation Network

BiLSTM Bidirectional Long Short-Term Memory

ADL Anti-Defamation League

SA Sturmabteilung

SS Schutzstaffel

NOP National Rebirth of Poland

YOLO You Only Look Once

ResNet Residual Network

SotA state of the art

NLP Natural Language Processing

KNN K-Nearest Neighbors

SVC Support Vector Classifier

SGD Stochastic Gradient Descent

BCE Binary cross-entropy

CCE categorical cross-entropy

NAdam Nesterov momentum variant of Adam

AUC-ROC Area Under the ROC Curve

TPR True Positive Rate

FPR False Positive Rate

ROC Receiver Operating Characteristic

AP Average Precision

PR Precision-Recall

AUC Area under the curve

API Application Programming Interface

JSON JavaScript Object Notation

OCR Optical character recognition

SHAP SHapley Additive exPlanations

Grad-CAM Gradient-weighted Class Activation Mapping

GDPR General Data Protection Regulation

PII personally identifiable information

DFG Deutsche Forschungsgemeinschaft

IEEE Institute of Electrical and Electronics Engineers

UNESCO United Nations Educational, Scientific and Cultural Organization

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